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Research paper

Discovery of novel *N*-acylpyrazoles as potent and selective thrombin inhibitors

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ABSTRACT

Direct oral anticoagulants (DOACs), which includes thrombin and factor Xa inhibitors, have emerged as the preferred therapeutics for thrombotic disorders, penetrating a market previously dominated by warfarin and heparin. This article describes the discovery and profiling of a novel series of *N*-acylpyrazoles, which act as selective, covalent, reversible, non-competitive inhibitors of thrombin. We describe *in vitro* stability issues associated with this chemotype and, importantly, demonstrate that *N*-acylpyrazoles successfully act *in vivo* as anticoagulants in basic thrombotic animal models. Crucially, this anticoagulant nature is unaccompanied by the higher bleeding risk profile that has become an undesirable characteristic of the DTIs and factor Xa inhibitors. We propose that the *N*-acylpyrazole chemotype shows intriguing promise as next-generation oral anticoagulants.

1. Introduction

1.1. Development of therapeutic agents against thrombotic disorders

Thrombosis and related disorders are severe, life-threatening conditions that represent a significant global challenge. The disease burden is large, chronic, and treatments are expensive [1].

Decades of research has led to a better understanding of the intricate and finely balanced coagulation process. Subsequent to an externally manifest injury, platelets are activated and then aggregate at the injury site [2]. Blood coagulation, tissue remodeling and then fibrinolysis leads to an overall physiological outcome of tissue healing. The coagulation cascade [3] includes the tightly controlled activation and deactivation of interconnected serine proteases, several of which are mediated by vitamin K as a critical co-factor. Identification of thrombin as a central serine protease in the coagulation cascade led to the structure-based discovery of putative inhibitors based upon endogenous substrates. Active site inhibitors that target the Ser-His-Asp catalytic triad led to the

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initial launch of argatroban in Japan in 1990 [4], and then FDA approval of dabigatran etexilate in 2010 [5]. Despite its poor bioavailability (5%), dabigatran etexilate has a better safety profile and wider therapeutic index than the unselective vitamin K antagonist warfarin through its more selective inhibition profile, resulting in a clinical situation where blood coagulation monitoring had been deemed unnecessary. However, as evidenced during extensive clinical experience, dabigatran is associated with undesired gastrointestinal bleeding. Subsequent to market launch, research *post hoc* has shown that dabigatran inhibits thrombin-induced platelet activation and aggregation [6]. This led to the development and launch of idarucizumab (Praxbind), an anticoagulant antidote [7], which is administered following uncontrolled bleeding episodes. While idarucizumab has been found to rapidly reverse the anticoagulant effect of dabigatran, frequently the patients suffering serious bleeding or needing emergency surgeries have additional comorbidities. Thus, it would be desirable for them to be anticoagulated with drugs with even lower bleeding risk profiles.

Similar drug discovery programs have pursued factor Xa (fXa), another serine protease central to the coagulation cascade. This has culminated in the launch of rivaroxaban, apixaban, edoxaban and betrixaban. These active site, competitive fXa inhibitors can also have associated uncontrolled bleeding risks and, as with the thrombin inhibitors, an anticoagulant antidote (e.g. andexanet alfa) was launched to combat these serious side-effects. More recent research has demonstrated that inhibition of other proteins in the coagulation cascade (e.g. fIXa, fXIa) also leads to anticoagulation, but crucially without the higher bleeding risks of the aforementioned fXa and direct thrombin inhibitors. Major drug discovery and development efforts against these protein targets are underway at several pharmaceutical companies, culminating in the preclinical and clinical disclosure of novel anticoagulants with safer bleeding profiles [8–10].

Other research has demonstrated that indirect (e.g. allosteric) thrombin inhibition results in anticoagulant efficacy [11,12]. Especially intriguing has been the possibility that undesired bleeding profiles characteristic of the active site, competitive, thrombin inhibitors, are avoided in allosteric inhibitors [13,14].

Lastly, *covalent*, active site serine protease inhibitors, considering their fundamentally distinct mechanisms of inhibition, will also likely display off-target effects distinct from those of the active site, competitive, thrombin inhibitors. The pharmacological profile of a novel, covalent serine protease inhibitor series will, by definition, be novel; this profile applies to both desired *and* undesired elements (i.e. 'on-target' and 'off-target' effects) of that series' pharmacology. The pursuit of any therapeutic discovery program will need to discern these pharmacological factors.

1.2. Covalent inhibition against serine protease targets

With covalent inhibition, the key binding event is the formation of a covalent bond between a protein-resident nucleophile and a small molecule drug. The success of the covalent inhibitor approach against cysteine-containing kinases [15,16] has prompted a similar approach against other nucleophile-containing enzymes. The serine hydrolase superfamily [17] encompasses a large class of important enzymes, with catalytic activity occurring through its active site-containing serine. As with inhibition of the kinases through covalent addition to an active site-resident thiol, a viable drug discovery practice has been the inhibition of the serine hydrolases *via* covalent addition to its active site-resident serine hydroxyl. A recent success story has been the discovery and development of telaprevir (Incivek™, Vertex/Janssen) [18], and boceprevir (Victrelis™, Merck) [19], both approved by the FDA in 2011 for the treatment of chronic hepatitis C. Both drugs act as competitive, yet covalent inhibitors of the decapeptide enzyme substrate NS5A/5B, acting through a reversible covalent linkage with the nucleophilic hydroxyl group on the Ser^{139} residue of NS3/4A protease [20].

1.3. Discovery of covalent inhibitors against thrombin

In undertaking a hit discovery, validation and optimization program, we were cognizant that decades of research and development had been conducted in the pursuit of competitive, reversible, active site thrombin inhibitors. Invariably, common starting 'hits' for many of these programs contained strongly basic guanidine and amidine P1 groups, which drive binding to thrombin's S1 specificity pocket [21]. In turn, developed 'products' arising from these hits were marketed drugs such as argatroban and dabigatran [22], subsequently known as direct thrombin inhibitors (i.e. DTIs, *vide supra*). As previously stated, in retrospect these DTIs are correlated with higher bleeding risks, leading us to avoid the initiation of a program based on reversible, competitive inhibitors. Instead, we took as inspiration multiple reports of covalent thrombin inhibition from even older literature, all shown in Fig. 1. For example, Lundblad *et. al.* reported the 'inactivation' of thrombin with *N*-acetylimidazole **1**, their interpretation being that several tyrosine residues of thrombin were acetylated by the activated reagent [23]. Tantalizingly, the same group subsequently reported that *N*-butyrylimidazole **2** acylated a residue at thrombin's 'active site', postulating that the enzyme's nucleophilic reactive center was either the serine hydroxyl group or the histidine imidazole unit [24]. Another group later reported thrombin's rapid inactivation by the *O*-activated oxazolone **3a** [25]; the now-inactivated enzyme could be reactivated after mild alkaline treatment, all indications consistent with reaction between a nucleophilic center resident in thrombin's active site and the activated reagent. Interestingly, the analog substituted at the oxazole 2-position with the methyl group (**3b**), rather than the phenyl group (**3a**), was much less active against thrombin. Additionally, *O*-activated oxazolone **3a** inactivated trypsin and chymotrypsin, but only slowly inactivated plasmin. These findings hinted at the possibility of enzyme selectivity by the design and synthesis of an appropriately adorned covalent inhibitor.

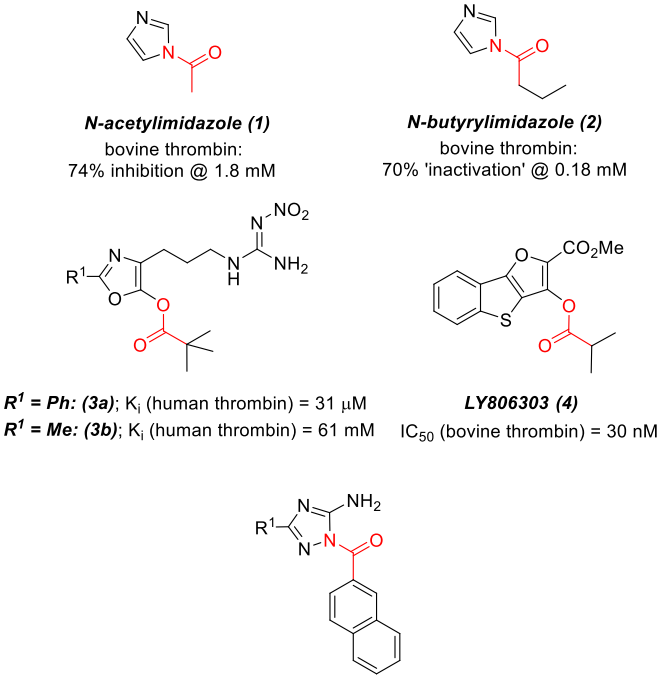


Fig. 1. Covalent thrombin inhibitors **1–5.**

Finally, Lilly Research Labs reported **4** (LY806303), a potent and selective thrombin inhibitor [26], once again shown to derive its activity *via* covalent inhibition. Similar to compounds **1**–**3**, *O*-activated benzothiophene-fused furanone **4** is primed for nucleophilic displacement at the acyl center. In this instance, the Lilly group proved by detailed mass spectrometric methods that thrombin had been isobutyrylated at its active site serine hydroxyl group [27]. To conclude, in the course of our program workers at the University of Münster reported a series of *N*-acyltriazoles (e.g. **5a, 5b**) that inhibit thrombin *in vitro*, also proving to be anticoagulants *in vivo* [28,29], thus adding support to this approach towards thrombin inhibition *via* covalent inhibition.

2. Results and discussion

2.1. Discovery and biochemical profiling of *N*-acyltriazole thrombin inhibitors

Consistent with the covalent inhibitors presented in Fig. 1, an initial collection of commercially available *N*-acylazoles of the general structure presented in Fig. 2 was insourced. Regioselective functionalization of the commercially available 3-amino-1,2,4-triazole 5-carboxylic acid provided the triazoles **6**–**10**; thus, chemoselective amidation of the exocyclic amine in **6** furnished amides **7** and **8**, whereas chemoselective amidation of the endocyclic N^1 nitrogen furnished amides **9** and **10.** Each compound was then screened for enzymatic inhibition against thrombin, fXa, and trypsin.

As shown in Table 1, amine **6** and exocyclic amides **7** and **8** displayed no measurable inhibition of any of the enzymes tested, whereas *N*-acyltriazoles **9** and **10** were verified to be potent inhibitors of thrombin. While we recognize that the reactive ester group at the triazole 5-position had the potential to drive thrombin reactivity, further investigations on compounds without this center would be required (*vide infra*).

Comparison of compounds **1–5** in Fig. 1 and compounds **6–10** of Table 1 suggested a path forward for designing novel covalent inhibitors of thrombin by investigating suitably substituted *N*-acyltriazoles (Fig. 2 (a)). A more general proposal was made to design and synthesize compounds based upon *N*-acylazoles, wherein the nucleophilic hydroxyl at Ser^{195} in thrombin's active site would displace the azole 'leaving group' *via* an addition-elimination reaction at the electrophilic acyl carbonyl

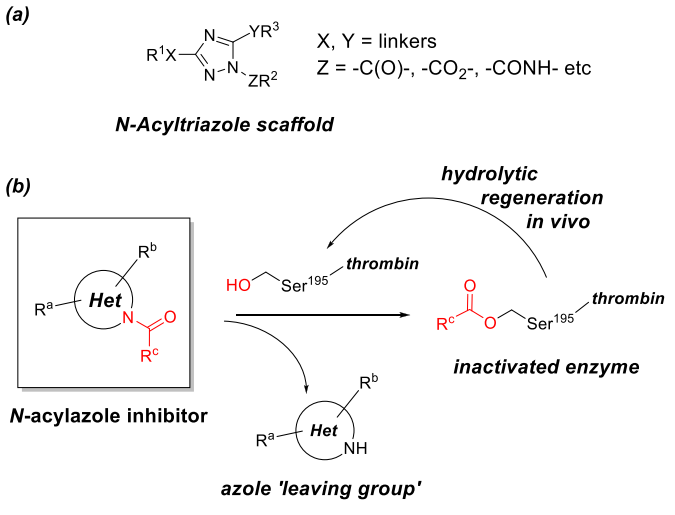


Fig. 2. (a); Proposed initial compounds based upon the *N*-acyltriazole scaffold. (b); Proposed design of novel covalent thrombin inhibitors.

Table 1
Enzymatic inhibition of derivatives **6-10**.

cmpd	Z	R^2	R^3	Enzymatic inhibition (IC_{50}, μM) Thrombin[a]	fXa[a]	Trypsin[b]
6	bond	H	H	>100	>100	>100
7	bond	H		>100	>100	>100
8	bond	H		>100	>100	>100
9	C(O)		H	0.09	>10	>10
10	C(O)		H	0.048	>10	>10

[a] Human.
[b] Bovine.

group. Thrombin would be temporarily esterified (i.e. inactivated), soon to be regenerated *in vivo* by hydrolytic means (Fig. 2(b)).

Encouraged by the series depicted in Table 1, we decided to employ an 'analog-by-catalog' approach, one of the fastest and most efficient ways to approach hit validation projects [30]. Following this strategy, we initially prioritized compounds for immediate synthesis and evaluation that were either available from commercial vendors or readily synthetically accessible *via* commercially available building blocks.

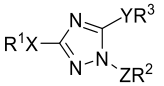
Compounds in Table 2 illustrate further exploration of the *N*-acyltriazole scaffold. Compound **13** [31] is one of many commercially available acylation products of symmetrical 3,5-diaminosubstituted 1,2,4-triazoles; however, this compound was inactive against all 3 serine proteases under our assay conditions. Interestingly, this chemotype has already shown utility elsewhere as kinase inhibitors [32–34].

Extension of the amine side chain (R^3) in the series previously described from Table 1 provided the benzylamino-substituted *N*-acyltriazole **11**. In this instance, there was no significant difference in thrombin inhibition vs. the free amine (IC_{50} (thrombin) = 55 nM (cmpd **11**) vs. IC_{50} (thrombin) = 48 nM (cmpd **10**)). However, this relative structure activity relationship was significantly different in compounds where the *N*-acyltriazole was *directly* substituted at the 3-carbon with an aromatic group. Thus, in comparison of the two 3-pyridyl-substituted *N*-acyltriazoles **14** and **15**, extension of the amine side chain at R^3 with a fluorobenzyl group dramatically improved thrombin inhibition: IC_{50} (thrombin) = 0.82 μM (cmpd **14**) [35] vs. IC_{50} (thrombin) = 3 nM (cmpd **15**), approximately a 300-fold increase in potency. We investigated other points of substitution, further narrowing our choice for an ideal chemotype. Switching the substitution at the 5-position of the *N*-acyltriazole from an amine to a thioether retained thrombin potency, but reduced selectivity vs. fXa and trypsin (compare compounds **15** and **16**). More dramatically, switching the nature of the $N^{triazole}$-acyl group completely abolished inhibition of all three proteins assayed. For example, carbamate **17** and urea **18** were inactive. These results are perhaps unsurprising in retrospect: fundamentally, effective covalent inhibition depends upon the cleavage ability of the $N^{triazole}$-acyl group on exposure to a thrombin-resident nucleophile. We can suppose that substitution of the $N^{triazole}$-acyl group with electron-rich O- or NH-groups hinders leaving group ability.

Finally, the stereoelectronic nature of the amine group at the *N*-acyltriazole 5-position was explored further. Replacement of the pendant proton with a methyl group dramatically reduced inhibition against thrombin: IC_{50} (thrombin) = 7 nM (cmpd **20**) vs. IC_{50} (thrombin) = 0.68 μM (cmpd **21**), approximately a 100-fold decrease in potency.

At this stage, based upon close inspection of ^{1}H NMR data for many analogs, we noted the intriguing possibility that the *N*-acyltriazole chemotype was present in the solution phase as an intramolecularly

Table 2
Enzymatic inhibition of *N*-acyltriazole derivatives **10-21**.

Cmpd							Enzymatic inhibition (IC_{50}, μM)		
	R^1	X	Z	R^2	Y	R^3	Thrombin[a]	fXa[a]	Trypsin[b]
10	Me	O_2C	C(O)		NH	H	0.048	>10	>10
11	Me	O_2C	C(O)		NH		0.055	>100	>100
12		bond	C(O)		NH		0.023	>100	>100
13		NH	C(O)		NH		>100	>100	n.d.
14		bond	C(O)		NH	H	0.82	>100	50
15		bond	C(O)		NH		0.003	22.3	87.7
16		bond	C(O)		S		0.005	2.38	0.218
17		bond	$C(O)_2$		NH		>100	>100	>100
18		bond	C(O)NH		NH		>100	>100	>100
19		bond	C(O)		NH		0.007	7.9	>100
20		bond	C(O)		NH		0.007	3.18	5.18
21		bond	C(O)		N(Me)		0.68	>100	13.2

[a] Human.
[b] Bovine.

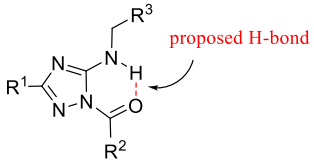


Fig. 3. Proposed presence of hydrogen bonding.

stabilized *H*-bonded structure (Fig. 3). *H*-Bonding had also been reported in similar structures by the group at the University of Münster, adding support to our observations [28,29,36,37]. We believe, without rigorous proof at this stage, that this *H*-bonding has utility, in both delivering the inhibitors to thrombin's active site in an ideal conformation for binding, and maximizing the chemotype's hydrolytic stability. This notion became important later in the program, when we paid closer attention to this *H*-bonding effect in other compounds.

With regard to mechanism of action, we wished to confirm that thrombin inhibition was derived primarily from the key covalent transacylation reaction between the Ser^{195} hydroxyl group in thrombin's active site and the labile $N^{triazole}$-acyl bond. Thus, we embarked on a two-pronged approach:

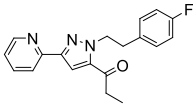
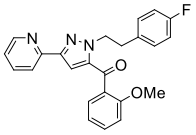
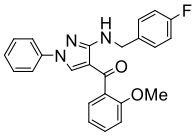
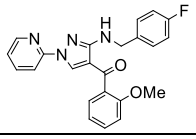
(i) synthesis and biochemical assay characterization of structural analogs *lacking* the scissile $N^{triazole}$-acyl bond;
(ii) biochemical characterization of thrombin inhibition with an appropriate *N*-acyltriazole.

2.1.1. Preparation of analogs lacking the scissile $N^{triazole}$-acyl bond

Several compounds structurally related to those in the *N*-acyltriazole hit series were synthesized, all designed to add further clarity to an emerging SAR (Table 3).

Firstly, potent thrombin activity in $N^{triazole}$-benzoyl compound **22** (IC_{50} (thrombin) = 10 nM) is consistent with that of the aforementioned *N*-acyltriazoles. However, thrombin activity was completely abolished in $N^{triazole}$-benzyl compound **23** (IC_{50} (thrombin) > 100 μM). Secondly, several isosteric compounds such as regioisomeric pyrazoles **24**–**27** were synthesized. Additional heterocyclic scaffold replacements were also made, such as oxazoles and thiazoles, but for the sake of brevity they are not described in this report. Each compound was designed to present the same substituent side chains in the aforementioned thrombin-active *N*-acyltriazole series, a strategic operation commonly referred to as 'scaffold hopping' [38]. However, the pyrazoles **24**–**27** lacked the scissile $N^{triazole}$-acyl bond, eliminating possible addition-elimination reactions with the hydroxyl group resident at the thrombin Ser^{195} active site. On the other hand, novel pyrazoles **24**–**27** contained an exocyclic ketone group as an isosteric replacement for the amide carbonyl center in the thrombin-active *N*-acyltriazoles. It was proposed that these compounds could take advantage of a reversible reaction between the thrombin Ser^{195} hydroxyl group and the ketone carbon i.e. a reversible formation of a hemiketal intermediate. This would be equivalent to an early approach taken by thrombin inhibitor research groups, wherein aldehyde scaffolds were designed as reversible 'serine traps' [39]. In any event, none of the pyrazoles **24**–**27** showed any hint of thrombin inhibition, again strengthening the working hypothesis that a predominant

Table 3
Enzymatic inhibition of derivatives **22-27**.

Cmpd #	R^1	R^2	R^3	Enzymatic inhibition (IC_{50}, μM) thrombin[a]	fXa[a]
22				0.01	28.3
23				>100	>100
24				>100	>100
25				>100	>100
26				>100	>100
27				>100	>100

[a] Human isoform.

factor in thrombin inhibition with the *N*-acyltriazole series is the scissile $N^{triazole}$-acyl bond.

2.1.2. Biochemical characterization

Initial kinetics experiments were conducted on the inhibition of thrombin and *N*-acyltriazole **19** (Fig. 4(a)). Human α-thrombin was pre-incubated with a vehicle solution of **19** at several concentrations, and reaction was started by the addition of the thrombin-specific chromogenic substrate (D)-FPR-*p*NA, also at several concentrations. The release rate of *p*-nitroaniline was recorded spectrophotometrically; the resultant Michaelis–Menten plot is shown in Fig. 4(b). The reciprocal plot i.e. the Lineweaver-Burk plot in Fig. 4(c), shows clearly that *N*-acyltriazole **19** inhibits α-thrombin in a non-competitive manner.

Further biochemical assays and binding kinetics were conducted in our laboratories, both on *N*-acyltriazole **19** and later compounds (data not reported in this article) [40,41]. Additionally, crystallographic experiments on a structurally related *N*-acyl***pyrazole*** are reported later in this manuscript (**84**, *vide infra*). All of the above confirms the compounds' covalent, non-competitive inhibitory behavior and adds further support and clarification for experimental results described in this article.

2.2. Development and evaluation of N-acyltriazole thrombin inhibitors

During work to further develop *N*-acyltriazoles as effective thrombin inhibitors, we recognized that the clinical efficacy of existing, competitive DTIs was correlated directly to functional efficacy, an easily obtained *in vitro* laboratory measurement. Functional efficacy of all the competitive, active site inhibitors developed thus far, both thrombin and fXa inhibitors, has commonly been measured by the activated partial thromboplastin time (aPTT) assay. Other functional assays have also been utilized, including prothrombin time (PT), thrombin time (TT) and

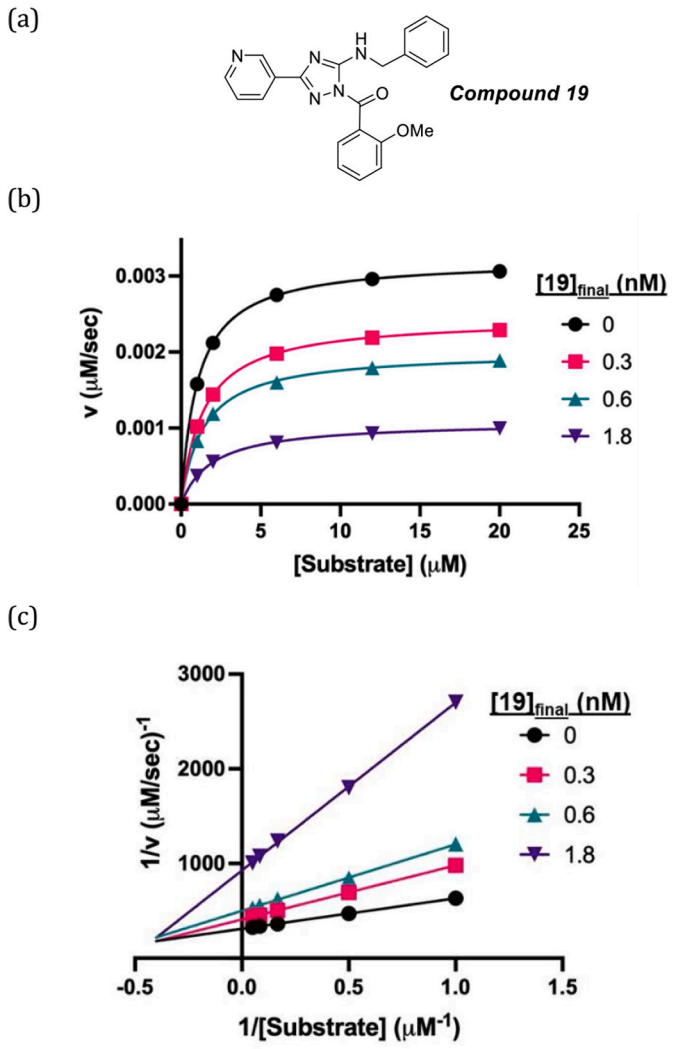


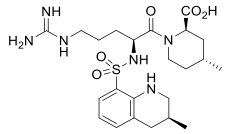
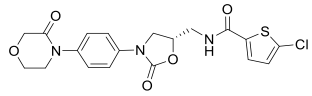
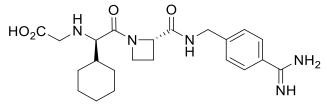
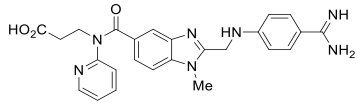
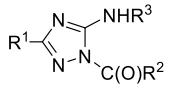
Fig. 4. (a); Compound **19**; (b); Michaelis–Menten plot of the inhibition of human α-thrombin with compound **19**; (c); Lineweaver-Burk plot of the inhibition of human α-thrombin with compound **19**.

ecarin clotting time (ECT), each of whose clotting functions are initiated by different agonists under different assay conditions. Several 'benchmark' compounds are shown in Table 4.

The aPTT assay is conducted in citrated normal pooled human platelet-poor plasma, and to it is then added the anticoagulant test compound, followed by assay reagent and then calcium chloride to trigger the clotting process [40]. Clot formation occurs in a matter of seconds. Functional efficacy of the anticoagulant test compound is defined by the concentration of test compound, which doubles the time for the clotting process. Ultimately, this CT_{2x} value is crucial: coupled with the compound's PK parameters, the CT_{2x} value dictates how large or how small will be the dose administered to the human subject. In other words, the smaller the aPTT CT_{2x} value, the better should be the anticoagulant development candidate going forward.

As shown in Table 4, the thrombin inhibition IC_{50} value of each 'benchmark' compound was on the same order of magnitude as its functional efficacy $CT_{2\times}$ value. For example, for argatroban: IC_{50} (thrombin) = 0.21 μM; vs. aPTT CT_{2x} = 0.27 μM. It was thus disconcerting to find that none of our *N*-acyltriazoles approached the submicromolar efficacy levels of the 'benchmark' compounds in Table 4; that is, 'efficacy' being defined by results of the aPTT assay. Our aPTT 'record holder' in the *N*-acyltriazole series is the propionamide **37** (Table 5) with an aPTT value over an order of magnitude inferior than

Table 4
Enzymatic inhibition and functional efficacy of benchmark DTIs, rivaroxaban, and novel *N*-acyltriazoles.

Cmpd name	Structure	Enzymatic inhibition (IC_{50}, μM)			Functional efficacy	
		Thrombin[a]	fXa[a]	Trypsin[b]	aPTT (CT_{2x}; μM)[c]	TGA (ETP; μM)[d]
Argatroban		0.21	71	59	0.27	0.36
Rivaroxaban		>100	0.006	>100	0.74	0.31
Melagatran		0.09	15.2	0.05	0.25	0.40
Dabigatran		0.04	5.8	1.23	0.11	0.3
		Enzymatic inhibition (IC_{50}, μM)			Functional efficacy	
Cmpd #	R^1	R^2	R^3	Thrombin[a]	fXa[a]	Trypsin[b]	aPTT (CT_{2x}; μM)[c]	TGA (ETP; μM)[d]
12				0.023	>100	>100	>100	18
19				0.007	7.9	>100	26	1.4

[a] Human.
[b] Bovine.
[c] CT, Clotting time.
[d] ETP, endogenous thrombin potential.

all the 'benchmark' thrombin inhibitors. Compare compound **37** (aPTT CT_{2x} = 11 μM) with argatroban (aPTT CT_{2x} = 0.27 μM).

In addition to the aPTT assay, our novel compounds were measured in the prothrombin time (PT) and the thrombin time (TT) assays. Methods descriptions of each are listed in the Experimental Procedures, and a full listing of these results is available in the Supporting Information.

The thrombin generation assay (TGA) has been refined and improved in the clinical environment in recent decades, addressing evident discrepancies and deficits associated with the aPTT and prothrombin time (PT) coagulation tests, and the derived international normalized ratio (INR), used widely in the clinic. Clinicians now understand that the TGA more accurately reflects the *actual* coagulation environment *in vivo*, than measurements made by the more traditional methods such as aPTT and PT assays [42]. Unlike the PT and aPTT assays, which use very high non-physiological agonist concentrations, the TGA is believed to use a physiologically relevant (~1 pM) amount of tissue factor. Moreover, the outputs of PT and aPTT reveal only the time at which clotting is macroscopically detected but provide no precise physical description of clot strength or composition. In contrast, the TGA output includes useful information such as lag time, peak height and area under the curve (AUC), termed 'endogenous thrombin potential' (ETP) [43], all of which correlate to clot strength and composition [44].

With the above observations in mind, considering that:

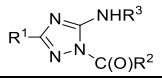
(a) we knew that our *N*-acyltriazole inhibitors inhibited thrombin in a non-competitive (and almost certainly covalent) manner, totally distinct from the known competitive 'benchmark' inhibitors (argatroban, melagatran, dabigatran), and since:
(b) the Thrombin Generation Assay (TGA) is possibly the most physiologically relevant assay for measuring inhibition of the coagulation cascade [41];

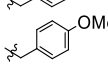
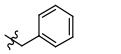
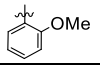
we decided to assess our *N*-acyltriazole thrombin inhibitors using the TGA while comparing with the 'benchmark' inhibitors [45].

Thus, the endogenous thrombin potential (ETP) for *N*-(*o*-anisoyl) triazole **19** was 1.4 μM in the TGA assay (Table 5), an order of magnitude 'more active' than its efficacy data as measured by the aPTT assay (CT_{2x} = 26 μM). Of course, the true 'efficacy' of our compounds could only be tested in thrombosis animal models, which we pursued in later compound series [40,41]. However, at this stage we felt confident that we were competitive with the 'benchmark' compounds shown at the top of Table 4. Even at this early stage of the project, we were able to discern some useful SAR (Table 5). It quickly became evident that efficacy was greatly influenced by the nature of the $N^{triazole}$-acyl group. While the *o*-chlorobenzoyl and electron-rich *o*-anisoyl groups are preferred, the electron-deficient *o*-cyanobenzoyl is not. For example, compare compound **30**: IC_{50} (thrombin) = 0.16 μM; ETP (TGA) > 100 μM; vs compound **28**: IC_{50} (thrombin) = 0.005 μM; ETP (TGA) = 2.7 μM. Worth mentioning here in the *N*-acyltriazole series are consistent discrepancies between potencies derived from biochemical assays (IC_{50}) and efficacies derived from TGA assays (EC_{50}), a difference of ~3 orders of magnitude between the two. This is in stark contrast to that of the 'benchmark' inhibitors (Tables 4 and 5). We believe there are valid reasons for these discrepancies related to PK properties; nevertheless, in developing an SAR we tracked both biochemical and efficacy data.

Dramatic effects were observed, on both enzymatic inhibition of thrombin and the TGA-measured efficacy, depending on the nature of the alkyl substituents of the $N^{triazole}$-acyl group. These are represented by compounds **36**–**41** in Table 5. For example, compare the propionamide **38** (IC_{50} (thrombin) = 0.028 μM; ETP (TGA) > 100 μM) with the isobutyramide **39** (IC_{50} (thrombin) = 0.007 μM; ETP (TGA) = 73 μM) and the pivalamide **40** (IC_{50} (thrombin) = 0.001 μM; ETP (TGA) = 1.8 μM). These shifts in thrombin inhibition are directly correlated with the level of alkyl substitution at the α-carbon (the carbon directly attached to the

Table 5
Enzymatic inhibition and functional efficacy of novel *N*-acyltriazoles.



cmpd #	R^1	R^2	R^3	Enzymatic inhibition (IC_{50}, μM)			Functional efficacy	
				Thrombin[a]	fXa[a]	Trypsin[b]	aPTT (CT_{2x}; μM)[c]	TGA (ETP; μM)[d]
12				0.023	>100	>100	>100	18
19				0.007	7.9	>100	26	1.4
22				0.01	28.3	14.2	>100	32
28				0.005	6.4	33	42	2.7
29				0.005	4.3	>100	34	31
30				0.16	6.9	>100	n.d.	>100
31				0.004	6.8	8.3	33	1.3
32				0.011	10.0	>100	>100	8.8
33				0.003	2.3	>100	>100	4.7
34				0.007	9.4	>100	n.d.	1.7
35				0.001	0.10	0.28	n.d.	23
36		Et		0.029	68	76	49	>100
37		Et		0.009	26	>100	11	>100
38		Et		0.028	39	>100	72	>100
39		^{i}Pr		0.007	>100	17.5	>100	73
40		^{t}Bu		0.001	>10	25.7	51	1.8
41		^{t}Bu		0.003	>10	>100	>100	>100
42	Me			>10	>10	n.d.	n.d.	n.d.
43	^{i}Pr			0.92	>100	>100	ND	>100
44				0.10	>100	14.7	>100	32

[a] Human.
[b] Bovine.
[c] CT, Clotting time.
[d] ETP, endogenous thrombin potential. n.d., not determined.

$N^{triazole}$-acyl group). ETP efficacy values are maximized in compounds where the α-carbon is exhaustively substituted with other alkyl groups. We found this trend replicated across other *N*-acyltriazole thrombin inhibitors (not shown). Although it is possible that the *tert*-butyl group represents the ideal group sterically, we speculate that maximal thrombin inhibition in the *tert*-butyl group is more related to the lack of available acidic protons α- to the $N^{triazole}$-carbonyl group. However, at this stage this theory remains untested. For now, we focused on the $N^{triazole}$-pivalamide group for much of the rest of the program, since compounds containing this group were imbued with ideal properties *in toto*.

With regard to the R^3 substituent, generally speaking the electron-rich anisyl-substituted amine side chain was not tolerated, although in the case of compound **41** this was only evident from the TGA result and not the thrombin enzyme inhibition assay. We explored the introduction of many other substituted aromatic side chains and heterocyclic side chains into the R^3 position (not shown in Table 5). One that became preferred was the R^3 = 5-(2′-chlorothienyl)methyl substituent, exemplified in compounds **33** and **35** in Table 5. We note that the α-chlorothienyl motif appears extensively in the serine protease drug discovery literature; the motif commonly protrudes deep into an S1 pocket when part of a competitive, active-site serine protease inhibitor [46], the most notable example being the fXa inhibitor rivaroxaban [47].

Compounds **32** and **34** compare well with compounds **12** and **31** respectively, inhibition against thrombin and TGA efficacy being approximately equivalent. Addition of the chlorine substituent at the α′-position of the thiophene substituent generally improved potency against thrombin even further; for example, compare compounds **32**

(IC_{50} (thrombin) = 0.011 μM; ETP (TGA) = 8.8 μM) with **33** (IC_{50} (thrombin) = 0.003 μM; ETP (TGA) = 4.7 μM).

However, this observational 'trend' was blemished somewhat in the comparison of the thiophene **34** with the α-chlorothiophene **35** (**34** (IC_{50} (thrombin) = 0.007 μM; ETP (TGA) = 1.7 μM) vs. **35** (IC_{50} (thrombin) = 0.001 μM; ETP (TGA) = 23 μM)). The dramatic down-shift in TGA efficacy for compound **35** is curious, since **35** is more potent in the inhibition of thrombin than compound **34**. However, in retrospect, given the general stability issues that soon became apparent in the entire *N*-acyltriazole thrombin inhibitor series (*vide infra*), it is possible that under the TGA assay conditions the *N*-acyltriazoles suffer competing hydrolytic decomposition.

Finally, changing the nature of the R^1 side chain also had a profound effect on thrombin enzyme inhibition and in the thrombin generation assay. The cyclohexyl-substituted *N*-acyltriazole **44** was much less potent and efficacious than the isosteric phenyl- and pyridyl-containing *N*-acyltriazoles, while the smaller methyl- (**42**) and isopropyl-substituted (**43**) *N*-acyltriazoles were effectively inactive. Simple aliphatic substitution at R^1 is not preferred.

2.3. Discovery and development of N-acylpyrazole thrombin inhibitors

In the optimization of *N*-acyltriazoles as efficacious thrombin inhibitors, not only were we cognizant of the need to be competitive with the 'benchmark' inhibitors as it pertains to functional efficacy, but we recognized the need for physicochemical profiling and subsequent development. Unfortunately, we were ultimately convinced that thrombin inhibitors containing the *N*-acyltriazole nucleus are inappropriate for further development, for stability reasons rather than efficacy reasons.

Until this point, we had noticed serious handling issues with all of the *N*-acyltriazoles. The most obvious of these was rapid degradation on exposure to CD_3OD (when measuring NMR spectra), or to methanol/water mixtures (when diluting samples for biological assays). Additionally, several compounds had earlier been assayed for buffer stability (PBS, pH 2, pH 8; results not shown in this publication). The sole side-product in every one of these instances (i.e. exposure to CD_3OD, aqueous buffer) was the non-acylated triazole, a process illustrated in Fig. 5(a).

Of course, this is the hydrolytic equivalent of the enzymatic active site transformation already depicted in Fig. 2(b). Separately, as part of our general screening paradigm, we subjected all of the *N*-acyltriazoles' immediate synthetic precursors (i.e. N–*H* triazoles) to thrombin inhibition assays and aPTT assays. Without exception, ***none*** of the N–*H* triazoles inhibited thrombin at a concentration level <100 μM (Fig. 5(b)). We recognize that in a subset of the *N*-acyltriazoles in Table 5 (e.g. **32**, **33**, **35**), their stabilities impact the interpretation of their potencies;

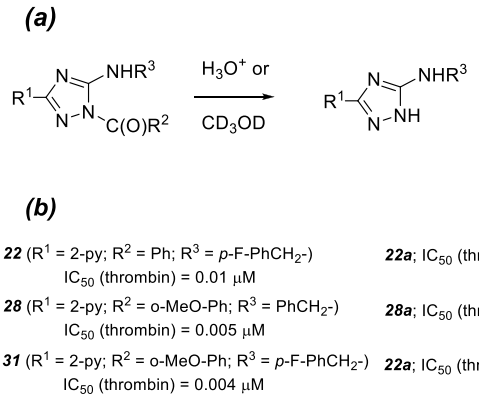


Fig. 5. (a); General hydrolysis reaction of *N*-acyltriazoles; (b); Thrombin inhibition data (IC_{50}) for a representative set of *N*-acyltriazoles and isolated N–*H* triazoles after hydrolysis, isolation and purification.

however, another subset of these compounds were both potent and stable, thus providing us sufficient SAR information. Nevertheless, in pursuit of more rigorous stability data, we ultimately instituted robust in-house *in vitro* plasma stability and liver microsome assays as a way to more conveniently assess compounds for further development.

Table 6 depicts plasma stability assay results across 3 species (CD-1 mouse; SD rat; human) for several *N*-acyltriazoles (**32**–**33**, **35**, **40**, **45**–**47**). Although this table is not fully populated, its data is sufficient enough to communicate a severe plasma instability when compared to that of argatroban. Whereas argatroban is stable in the presence of all three plasma species, each of the *N*-acyltriazoles in Table 6 degrades significantly on exposure to each at least one of the plasma species tested.

On the other hand, despite a general trend for lower potency in the thrombin enzyme assay, we were greatly encouraged by examples of significantly more plasma-stable *N*-acylpyrazoles (**48**, **49** in Table 6), especially given the trajectory required for drug development and the desire for optimal properties *in vivo* [48,49]. Thus, we were motivated to focus our attention on thrombin inhibitors containing the *N*-acylpyrazole motif. Guided by the SAR gathered during the earlier synthesis and assay of *N*-acyltriazoles, we embarked on a systematic hit identification campaign of *N*-acylpyrazoles with emphasis on varying the three side chains $R^{1\text{-}3}$. Our goal was to discover compounds with low single-digit micromolar TGA efficacy, for further evaluation and development.

With the efficacy of *N*-acyltriazole **40** and the plasma stability of *N*-acylpyrazole **49** as reasonable starting points, compounds **49**–**54** (Table 7) were synthesized with the side chains R^3 = (*p*-fluorophenyl) methyl and *N*-pivaloyl groups invariant. Pair-wise comparison of compounds **49** and **50** revealed the requirement for an *H*-bond acceptor within the R^1 side chain (compare *N*-acylpyrazole **49** (IC_{50} (thrombin) = 0.047 μM) with *N*-acylpyrazole **50** (IC_{50} (thrombin) = 10.9 μM)). Pair-wise comparison of compounds **49** and **51** revealed the importance of the *location* of the required *H*-bond acceptor i.e. a sole heteroatom in the R^1 *meta* position was not preferred (compare *N*-acylpyrazole **51** (IC_{50} (thrombin) = 2.2 μM) with *N*-acylpyrazole **49** (IC_{50} (thrombin) = 0.047 μM)). From this closed set of compounds, the 2,3-pyridazine **54** was the most promising candidate (IC_{50} (thrombin) = 0.005 μM; ETP (TGA) = 6.9 μM).

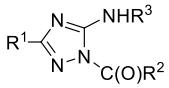
Compounds **49**, **55**–**60** represents a series of compounds that explored the *N*-acylpyrazole R^3 group. Compounds in this set held invariant the R^1 = 2-pyridyl and the *N*-pivaloyl groups, with the exception of 2,3-pyridazine **59**. We explored a series of bioisosteric replacements for the phenyl group, the most effective being the 2-thienyl group (compare **55** (R^3 = phenylmethyl): IC_{50} (thrombin) = 0.07 μM; ETP (TGA) = 54 μM) with **57** (R^3 = (2-thienyl)methyl): IC_{50} (thrombin) = 0.04 μM; ETP (TGA) = 32 μM)). Finally, both thrombin activity and TGA efficacy improved noticeably with the introduction of a chlorine substituent onto the thiophene ring at the α′-position (compare **60** (R^3 = (2-(5′-chlorothienyl)methyl): IC_{50} (thrombin) = 0.008 μM; ETP (TGA) = 11 μM) with **57** (R^3 = (2-thienyl)methyl): IC_{50} (thrombin) = 0.04 μM; ETP (TGA) = 32 μM)). This trend compares well with the increased thrombin potency for *N*-acyltriazoles **33** and **35** (*vide supra*).

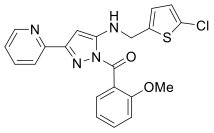
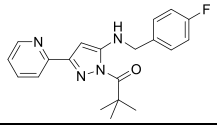
Table 8 depicts a series of *N*-acylpyrazoles in which the acyl group was varied, keeping invariant the preferred groups R^1 = 2-pyridyl and R^3 = 2-(5′-chlorothienyl)methyl. Furthermore, Table 8 is subdivided into a group of compounds whose *N*-acyl group is aromatic (**48**, **61**–**67**), and a group of compounds whose *N*-acyl group is aliphatic (**60**, **68**–**72**).

With the exception of extended substitution (e.g. in *o*-phenoxybenzoyl derivative **67**) the R^2 aromatic group was broadly tolerated. Compounds containing the benzoyl isosteres 3-furoyl (**62**) and 3-thienoyl (**63**) were somewhat comparable with the *N*-benzoyl derivative **66** (compare **62** (R^2 = 3-furyl): IC_{50} (thrombin) = 0.009 μM; ETP (TGA) = 13 μM with **63** (R^2 = 3-thienyl): IC_{50} (thrombin) = 0.006 μM; ETP (TGA) = 8.2 μM) and **66** (R^2 = phenyl): IC_{50} (thrombin) = 0.005 μM; ETP (TGA) = 4.6 μM. The *o*-aminobenzoyl derivative **65** was equipotent with the parent *N*-benzoyl derivative **66**, appearing to demonstrate in

Table 6
Enzymatic inhibition, functional efficacy and plasma stability of argatroban and novel *N*-acylazoles.

Compound	Thrombin inhibition (IC_{50}; μM)[a]	TGA (ETP; μM)[b]	Plasma stability (CD-1 mouse; $t_{½}$ (min))	Plasma stability (SD rat; $t_{½}$ (min))	Plasma stability (human; $t_{½}$ (min))
Argatroban	0.21	0.36	>120	>120	>120



cmpd #	R^1	R^2	R^3	Thrombin inhibition (IC_{50}; μM)[a]	TGA (ETP; μM)[b]	Plasma stability (CD-1 mouse; $t_{½}$ (min))	Plasma stability (SD rat; $t_{½}$ (min))	Plasma stability (human; $t_{½}$ (min))
32				0.011	10.0	n.d.	2.2	>120
33				0.003	4.7	16	n.d.	n.d.
35				0.001	23	4.1	1.5	30
40		^{t}Bu		0.001	1.8	n.d.	4.7	12
45		Et		0.009	>100	n.d.	7.8	19
46				0.008	8.3	n.d.	n.d.	48
47				0.02	24	n.d.	n.d.	69
48				0.005	4.1	81	35	130
49				0.047	42	750	>120	>120

[a] Human.
[b] ETP, endogenous thrombin potential. n.d., not determined.

this limited set a lack of preference for electron-rich or electron-poor acyl substituents (compare **65** (R^2 = aminophenyl): IC_{50} (thrombin) = 0.004 μM; ETP (TGA) = 5.4 μM with **66** (R^2 = phenyl): IC_{50} (thrombin) = 0.005 μM; ETP (TGA) = 4.6 μM). Additional electron-donating groups, such as the acyl groups present in *N*-acylpyrazoles **61** and **64**, did not improve potency or efficacy beyond the parent *N*-benzoyl derivative **66**. However, addition of substantial steric substitution was deleterious: the *o*-phenoxybenzoyl derivative **67** displayed significantly diminished thrombin inhibition and was ineffective in the TGA.

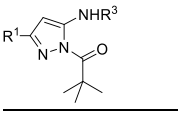
In contrast to the thrombin inhibition SAR displayed by the *N*-benzoyl and *N*-heteroaroyl derivatives (**48**, **61**–**67**), the SAR of the aliphatic-substituted *N*-acylpyrazoles (**60**, **68**–**72**) is more noteworthy. Addition of a polar hydroxyl group at the terminus of the pendant pivaloyl group markedly increased potency against thrombin, and increased efficacy in the thrombin generation assay (TGA) (compare **60** (R^2 = *tert*-butyl: IC_{50} (thrombin) = 0.008 μM; ETP (TGA) = 11 μM) with **68** (R^2 = 2-hydroxy-1,1-dimethylethyl: IC_{50} (thrombin) = 0.001 μM; ETP (TGA) = 3.4 μM)). The requirement that the carbon α' to the *N*-acyl group be quaternary (*vide supra*) was again illustrated in the contrast between the two tetrahydropyran derivatives **70** and **71** (compare **70** (IC_{50} (thrombin) = 0.174 μM; ETP (TGA) > 100 μM) with **71** (IC_{50} (thrombin) = 0.003 μM; ETP (TGA) = 8.6 μM)). Finally, addition of more lipophilic content dramatically diminished thrombin inhibition and efficacy (compare the *N*-pivaloyl derivative **60** (IC_{50} (thrombin) = 0.008 μM; ETP (TGA) = 11 μM) with the α-(methyl)cyclohexyl derivative **72** (IC_{50} (thrombin) = 0.177 μM; ETP (TGA) > 100 μM)).

From the discussion above, to achieve potent thrombin inhibition it appeared critically important that *N*-acylpyrazoles contain an *H*-bond acceptor in the R^1 substituent adjacent to the connection to the pyrazole nucleus. As a review, compare derivatives **49**–**54** in Table 7. Additionally, it should be noted that until now very few compounds have been described where R^1 is aliphatic rather than aromatic, with compounds **42**–**44** in Table 5 as the sole exceptions (albeit examples of *N*-acyltriazoles). We had also observed greater chemical stability in these and *N*-acylpyrazoles containing aliphatic R^1 groups; these observations associated with favorable substituents in *N*-acylpyrazole **60** prompted us to explore SAR optimizing efficacy and stability. With this in mind, the following describes *in vitro* stability profiles of key *N*-acylpyrazoles, with the first described *in vivo* efficacy and PK data.

Most compounds with promising *in vitro* thrombin potency and efficacy in the thrombin generation assay were profiled in plasma stability assays, beginning with that of the CD-1 mouse and including other species where necessary. As part of the workflow, compounds were routinely submitted to *in vitro* liver microsome stability assays [49]; again, microsomal extracts from the CD-1 mouse were used primarily, followed by other species' extracts where necessary.

From Table 9, 2-pyridyl derivative **60** exhibited good stability ($t_{½}$ = 360 min) when exposed to CD-1 mouse plasma, but unfortunately suffered rapid clearance on exposure to CD-1 mouse liver microsomes (Cl_{int} = 1500 μL/min/mg). Addition of a 2nd nitrogen into the R^1 ring afforded the 2,3-pyridazine derivative **59**, which increased potency in the thrombin inhibition assay, and improved efficacy in the TGA; compare **60** (IC_{50} (thrombin) = 0.008 μM; ETP (TGA) = 11 μM) with **59** (IC_{50} (thrombin) = 0.002 μM; ETP (TGA) = 4.0 μM). This improvement

Table 7
Enzymatic thrombin inhibition and functional efficacy of *N*-pivaloylpyrazoles.

Cmpd #			Thrombin inhibition (IC_{50}; μM)[a]	TGA (ETP; μM)[b]	Ratio of TGA ETP/thrombin IC_{50}
	R^1	R^3			
49			0.047	42	893
50			10.9	n.d.	–
51			2.2	n.d.	–
52			0.024	49	2042
53			0.025	31	1240
54			0.005	6.9	1380
55			0.07	54	771
56			0.23	>100	–
57			0.04	32	800
58			0.13	73	562
59			0.002	4.0	2000
60			0.008	11	1375

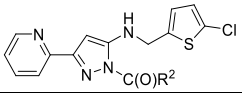
[a] Human.
[b] ETP, endogenous thrombin potential. n.d., not determined.

mirrors the same trend reported in Table 7 with the 2-pyridyl derivative **49** and the 2,3-pyridazine **54** derivative. On exposure to BALB/c mouse microsomes, 2,3-pyridazine **59** also suffered high clearance (Cl_{int} = 200 μL/min/mg). This compound was also assayed against microsomes from a variety of other species (e.g. guinea pig; beagle dog; cynomolgous monkey), with equally rapid clearance. In contrast, clearance was quite low with this compound on exposure to human liver microsomes (Cl_{int} = 6 μL/min/mg). Similarly, clearance in mouse plasma for pyrrolidine derivative **76** was disappointingly high (Cl_{int} (CD-1 mouse) = 200 μL/min/mg), whereas clearance in human plasma was quite acceptable (Cl_{int} (human) = 7 μL/min/mg). These results are encouraging, but still require compounds stable in low preclinical species to comply with FDA guidance: a development compound must have *in vitro* plasma- and microsomal stability data in at least two species, to be predictable and correlative with stability in human subjects in the clinical trial setting [50].

The 3-pyridone **73** (Table 9) represents an interesting development: the 3-pyridone moiety incorporates the required *H*-bond acceptor (*vide supra*), and indeed this compound was quite potent and efficacious (IC_{50} (thrombin) = 0.001 μM; ETP (TGA) = 1.8 μM). While reasonably stable on incubation with CD-1 mouse plasma, 3-pyridone **73** was significantly more stable on incubation with CD-1 mouse and human microsomes (Cl_{int} (mouse) = 60 μL/min/mg; Cl_{int} (human) = 17 μL/min/mg. Regioisomeric pyridones **74** and **75** offered no significant advantage in TGA efficacy over pyridone **73** (compare **73** (ETP (TGA) = 1.8 μM) with **74** (ETP (TGA) = 6.8 μM) and **75** (ETP (TGA) > 100 μM)).

In order to determine oral bioavailability, *N*-acylpyrazole **73** was orally dosed in male CD-1 mice (5 mg/kg, formulated in 90% solutol HS15, 10% ethanol) and pharmacokinetics tracked for a period of 24 h, as previously described by our laboratories [40]. Fig. 6 shows that AUC and C_{max} were 713.3 (±594.5) h.ng/mL and 127 (±117.4) ng/mL respectively. Half-life ($t_{½}$) of *N*-acylpyrazole **73** after oral dosing was 5.9 (±2.7) hours, and theoretical bioavailability (%F) was 22.8 (±16.7).

Table 8
Enzymatic thrombin inhibition and functional efficacy of *N*-pivaloyl-3-(2-pyridyl)pyrazoles.



Cmpd #	R^2	Thrombin inhibition (IC_{50}; μM)[a]	TGA (ETP; μM)[b]
48		0.005	4.1
61		0.012	6.9
62		0.009	13
63		0.006	8.2
64		0.01	11
65		0.004	5.4
66		0.005	4.6
67		0.034	>100
60	^{t}Bu	0.008	11
68		0.001	3.4
69		0.002	11
70		0.174	>100
71		0.003	8.6
72		0.177	>100

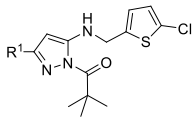
[a] Human.
[b] ETP, endogenous thrombin potential.

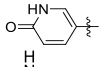
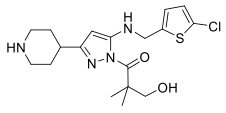
Further interest in this latter set of compounds came from their *in vitro* stability and *in vivo* pharmacokinetics profiles (Table 9). *In vitro* liver microsomal clearance in the CD-1 mouse was reduced to quite acceptable levels in the 4-piperidines **77**, **78** and **82** (**77**: Cl_{int} (CD-1 mouse) = 11 μL/min/mg; **78**: Cl_{int} (CD-1 mouse) = 21 μL/min/mg; **82**: Cl_{int} (CD-1 mouse) = 19 μL/min/mg).

The relatively long i.v. half-life ($t_{½}$ = 3.6 h) for piperidine **77** in the CD-1 mouse was also intriguing, and hinted at a promising development path forward for future compounds, despite its relatively poor overall exposure (AUC = 780 h.ng/mL).

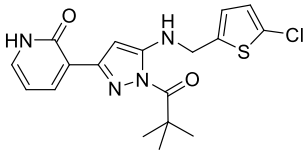
To complete a systematic SAR exploration, and as a way of probing *in vitro* potency, efficacy and stability of the *N*-acylpyrazole nucleus, we prepared several analogs with fluorine substitution at the pyrazole 4-position. Table 10 depicts several pairs of *N*-acylpyrazoles and *N*-acyl(4-fluoropyrazoles), keeping invariant R^3 = 2-(5′-chlorothienyl)methyl as the preferred group. In every instance, addition of fluorine in the pyrazole 4-position boosted inhibition against thrombin *in vitro*, consistent with the anticipated increased leaving group ability of the azole (cf. Fig. 2(b)). As with previous examples, increased potency in the thrombin enzyme assay translated to increased efficacy in the thrombin generation assay (TGA). However, fluorine substitution consistently imparted decreased stability against all plasma species studied. For example,

Table 9
Enzymatic thrombin inhibition, functional efficacy, *in vitro* and *in vivo* stability of *N*-pivaloylpyrazoles.



cmpd #	R^1	*In vitro* activity			*In vitro* stability[a]		*In vivo* PK	
		Thrombin inhibition (IC_{50}; μM)[b]	TGA (ETP; μM)[e]	Ratio of TGA ETP/ thrombin IC_{50}	Plasma (CD-1 mouse, $t_{½}$ (min))	LM (CD-1 mouse, CL_{int} (μL/min/mg))	PK, CD-1 mouse (i.v. $t_{½}$ (h))	PK, CD-1 mouse (i.v. AUC (h•ng/mL))
60		0.008	11	1375	360	1500	0.2	280
59		0.002	4.0	2000	51[b]	200[c]; 6[b]	n.d.[f]	n.d
73		0.001	1.8	1800	350	60; 17[b]	0.7[d]	300[d]
74		0.001	6.8	6800	50	53	n.d.[f]	n.d.[f]
75		0.19	>100	–	190	38	n.d.[f]	n.d.[f]
76		0.019	5.1	268	200	200; 7[b]	1.0	83
77		0.21	26	124	300	11	3.6	780
78		0.76	18	24	1100	21	n.d.[f]	n.d.[f]
79		0.03	6.4	213	910	110	n.d.[f]	n.d.[f]
80		0.012	11.0	917	n.d.[f]	100[b]	n.d.[f]	n.d.[f]
81		0.36	>100	–	>300	110	n.d.[f]	n.d.[f]
82		0.21	8.5	40	320	19	1.3	230

[a] In given species, unless denoted otherwise.
[b] Human.
[c] BALB/c mouse.
[d] Dosed at 0.5 mg/kg.
[e] ETP, endogenous thrombin potential.
[f] n.d., not determined.

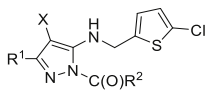


73

p.o. AUC (h.ng/ml)	***713.3 (± 594.5)***
p.o. Cmax (ng/ml)	***127 (± 117.4)***
p.o. Tmax (h)	***0.7 (± 0.3)***
p.o. t½ (h)	***5.9 (± 2.7)***
%F	***22.8 (± 16.7)***

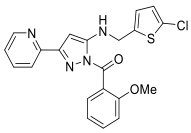
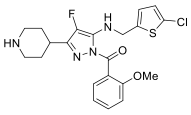
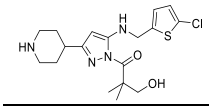
Fig. 6. Pharmacokinetics parameters and calculated bioavailability (%F) of *N*-acylpyrazole **73** in the CD-1 mouse.

Table 10
Enzymatic thrombin inhibition, functional efficacy, and *in vitro* plasma stability of *N*-acylpyrazoles and *N*-acyl(4-fluoropyrazoles).

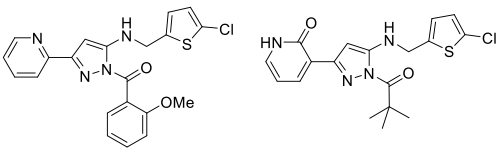
cmpd #				*In vitro* activity		*In vitro* plasma stability
	R^1	R^2	X	Thrombin inhibition (IC_{50}; μM)[a]	TGA (ETP; μM)[b]	CD-1 mouse ($t_{½}$, min)
83			H	1.2	22	43
84			F	0.06	4.1	8.4
77		^{t}Bu	H	0.21	26	700
85			F	0.01	2.8	72
82			H	0.21	8.5	190
86			F	0.03	3.4	44
87			H	1.1	>100	160
88			F	0.044	30	12
81		^{t}Bu	H	0.36	>100	>300
89			F	0.008	46	240
90			H	0.034	2.2	310
91			F	0.004	2.3	50

[a] Human.
[b] ETP, endogenous thrombin potential.

Table 11
Inhibition of platelet activation by standard-of-care thrombin- and fXa inhibitors and *N*-acylpyrazoles **48**, **82**, **84**.

Compound	#	Platelet activation, (EC_{50}, μM)[a]
Argatroban		0.007; 0.024[b]
Dabigatran		0.007; 0.029[b]
Rivaroxaban		>10; 0.01[b]
Apixaban		0.07[b]
	**48**	4.4
	**84**	11
	**82**	43

[a] Unless otherwise indicated, assays were conducted in PRP, CD-1 mouse, thrombin as agonist.
[b] Conducted in 76% human blood, with TF activation.



Serine protease	48	73
Thrombin	0.005	0.001
fVIIa	>100	>100
fIXa	>100	>100
fXa	>10	>100
fXIa	9.2	76
fXIIa	>100	>100
aPC	>100	>100
Chymotrypsin	0.67	0.38
Plasmin	4.9	>100
Trypsin	>100	>100
Tryptase	>100	>100
Plasma kallikrein	1.8	20.1

Enzymatic inhibition (IC_{50}, μM)

Fig. 7. *N*-Acylpyrazoles **48** and **73** and their enzymatic activities against a panel of serine proteases.

compare *N*-acylpyrazole **83** (IC_{50} (thrombin) = 1.2 μM; ETP (TGA) = 22 μM; $t_{½}$ (CD-1 mouse plasma) = 43 min) with *N*-acyl(4-fluoropyrazole) **84** [40] (IC_{50} (thrombin) = 0.06 μM; ETP (TGA) = 4.1 μM; $t_{½}$ (CD-1 mouse plasma) = 8.4 min). Only plasma stability in a single species (CD-1 mouse) is shown in Table 10 for the sake of brevity.

Crucially, we disclose this novel scaffold's off-target effects. Table 11 shows a small selection of compounds and their inhibition of platelet activation. Argatroban and dabigatran (selective thrombin inhibitors) are very potent inhibitors of platelet activation (EC_{50} (CD-1 mouse) = 7 nM), while rivaroxaban and apixaban (selective fXa inhibitors) also potently inhibit the activation of platelets (rivaroxaban EC_{50} (human) = 10 nM; apixaban EC_{50} (human) = 70 nM). On the other hand, novel *N*-acylpyrazole thrombin inhibitors are only weak inhibitors of platelet activation. For example, the inhibition of platelet activation by *N*-acyl(4-fluoropyrazole) **84** [40] in the CD-1 mouse is EC_{50} = 11 μM, 3 orders of magnitude less potent than all the anticoagulants currently available as standards of care. This lack of inhibition of platelet activation has previously been disclosed by our laboratories for a compound containing the same *N*-acylpyrazole chemotype, and whose biochemical properties were more extensively characterized than compounds in this report [41]. We illustrate this *in vitro* profile in conjunction with the *in vivo* phenomenon described in the introduction of this manuscript: that subsequent to an externally manifest injury, platelets are activated and then aggregate at the injury site [2]. We speculate that the lack of this off-target effect in our novel *N*-acylpyrazoles, and the clinical observation that many of the anticoagulant standards-of-care are potent inhibitors of platelet activation, present our novel *N*-acylpyrazoles with a unique opportunity whose *in vivo* characterization will be described in the following section (*vide infra*).

Importantly, regarding selectivity against functionally related proteases, Fig. 7 illustrates two representative *N*-acylpyrazoles and their *in vitro* inhibition of thrombin and a panel of relevant anti-targets. Overall, this selectivity profile was also apparent in other compounds from this series, and so we felt confident that these limited off-target effects would be apparent for compounds in further development work.

Finally in this *in vitro* section, we report enlightening results derived from X-ray diffraction experiments. Many earlier attempts had been made to co-crystallize *N*-acyltriazoles with thrombin, without success. However, in due course crystals were obtained from the incubation of *N*-acylpyrazole **84** with human α-thrombin. After data collection and structural refinement, two thrombin molecules were found in each asymmetric unit (Fig. 8). Electron density analysis showed that in both monomers, both catalytic serine residues had been modified. Additionally, the only 'new' electron density did not correspond to the intact *N*-acylpyrazole **84**, but instead an *o*-methoxybenzoyl unit adjacent to the catalytic Ser^{195} hydroxyl group.

As shown in Fig. 8(a), electron density correlates to a new methoxybenzene unit and a carbonyl oxygen adjacent to the thrombin Ser^{195} hydroxyl group. Fig. 8(b) also shows a monomer with the *o*-methoxybenzoyl unit, and continuous electron density consistent with a covalent linkage to the thrombin Ser^{195} hydroxyl group. The non-ideal quality of the structure (i.e. that there were two monomers per unit cell) is not an unexpected result as the crystallization process occurred over five days. Kinetics experiments [40] determined that hydrolysis of the acylated thrombin occurs with a half-life of $t_{½}$ ~ 8 h, making this process competitive with the length of the crystallization experiment itself, and hence explaining the disordered structures in Fig. 8 [40]. All the above is consistent with the formation, crystal growth and isolation of the 'inactivated enzyme' (R^c = *o*-MeO-Ph) depicted in Fig. 2(b). The experimental coordinates have subsequently been deposited to the PDB as code 6CYM [40,51]. A full description is given in the Supporting Information.

Taking the enzymatic selectivity data in Table 5, together with the superior efficacy (TGA) for 3-pyridone **73** and its exposure profile in mouse matrices, *N*-acylpyrazole **73** was profiled further in efficacy evaluation studies, as follows.

2.4. In vivo evaluation of N-acylpyrazole thrombin inhibitors

2.4.1. In vivo efficacy evaluation

In order to understand the effect of *N*-acylpyrazole **73** *in vivo*, we sought to correlate its concentrations *in vivo* and *in vitro*. Fig. 9(a) shows the *in vitro* thrombin generation assay (TGA) curve for *N*-acylpyrazole **73**. As already illustrated in Table 9, this compound's ETP EC_{50} = 1.8 μM. However, to approximate complete suppression of thrombin generation *in vivo*, we sought to achieve an inhibitor concentration at its EC_{98}. From Fig. 9(a), this is approximately 33 μM. Fig. 9(b) depicts key data from which the concentration of *N*-acylpyrazole **73** at its EC_{98} *in vitro* was used to estimate its 'ideal' concentration in an *in vivo* efficacy test; i.e. the thrombin-induced thromboembolism model, *vide infra*.

Assuming linearity, we derived the i.v. dose for the *in vivo* efficacy test from the following equation:

$$\text{i.v. dose in an efficacy test } (\text{mg}/\text{kg}) = \frac{(\text{target i.v. } C_0 \text{ in efficacy test } (\text{nM})) \times (\text{i.v. PK dose } (\text{mg}/\text{kg})) \times (\text{molecular weight } (\text{g}/\text{mol}))}{\text{IV PK } C_0 \ (\text{ng}/\text{ml}) \times 1000}$$

From this equation, we derived the i.v. dose for the efficacy test to be 8.7 mg/kg. For ease of application however, the following thrombin-induced thromboembolism *in vivo* model study was conducted using an i.v. dose of 9 mg/kg.

The thrombin-induced thromboembolism model [52] has been developed to further investigate the role of endogenous proteins (e.g.

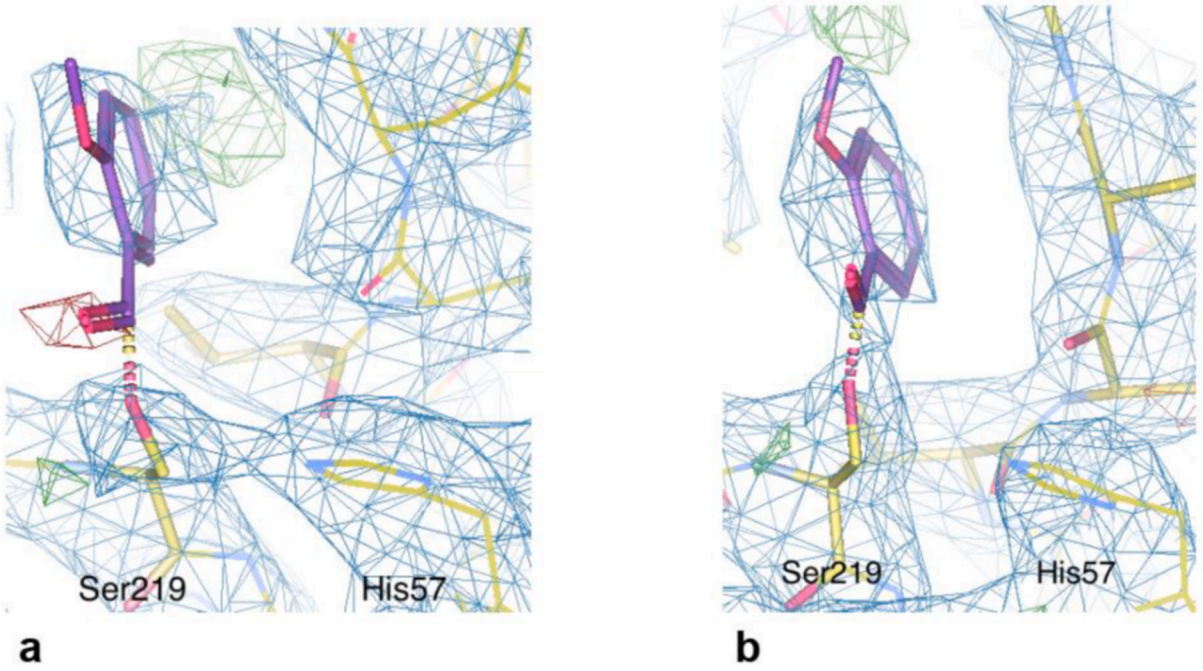


Fig. 8. Refined structure of human α-thrombin with *o*-methoxybenzoyl covalently bound to Ser^{219} (Ser^{195} in the thrombin nomenclature); a); Blue electron density map contoured at 1σ around Ser^{219} (Ser^{195} in the thrombin nomenclature) covalently linked (dots) to *o*-methoxybenzoyl (magenta) in chain A. b); Blue electron density contoured at 1σ around Ser^{219} linked (dots) to *o*-methoxybenzoyl (magenta) in chain C. For clarity the hydrogen bonds have been omitted. Traces of electron difference density (Fo-Fc) map contoured at 3σ are shown in red (negative) and green (positive).

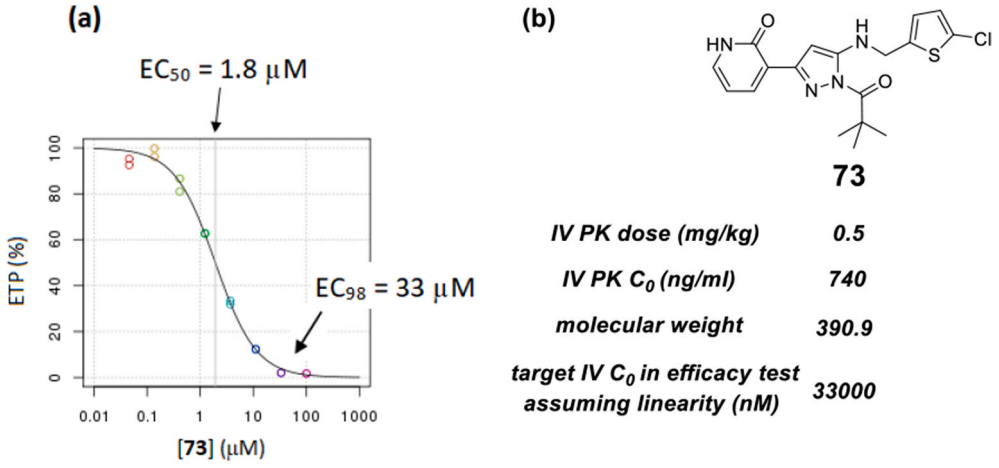
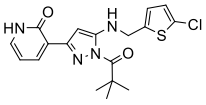


Fig. 9. (a); *In vitro* thrombin generation assay (TGA) curve for *N*-acylpyrazole **73**. ETP = endogenous thrombin potential. (b); Data derived from the same PK studies reported in Table 9 for *N*-acylpyrazole **73.**

Table 12
Thrombin-induced thromboembolism study of argatroban and *N*-acylpyrazole 73.

Study article	Thrombin-induced thromboembolism study			
	dosage (mg/kg)[a]	# mice in study	# survived mice	mice survival %
Vehicle[b]		4	0	0
Argatroban	8	6	6	100
	4	6	4	67
	2	4	0	0
 **73**	**9**	**7**	**7**	**100**

[a] Intravenous (i.v.) administration of test article (mg) per mouse weight (kg).
[b] DMA:PEG300:phosphate buffer (5:1:4), plus 20 μM β-cyclodextrin.

aPC) [53] in the coagulation cascade, and to assess the anticoagulant effect of novel therapeutics (e.g. low molecular weight heparins) [54]. In this model, mice challenged with bolus-administered thrombin suffer rapid mortality through acute fibrin deposition. However, either aPC or therapeutic administration of an anticoagulant prior to thrombin challenge protects the mice in a dose-dependent manner. Although massive thrombin dosage in this *in vivo* model is considered artificial and not representative of the clinical setting, it does provide a rapid sense of overall effectiveness of anticoagulant efficacy in a novel therapeutic.

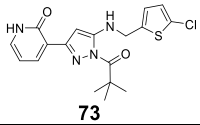
To that end, several study groups of CD-1 mice (male, 25–30 g) were acquired (Harlan Labs, Livermore, CA), prepared and handled according to AAALAC ethical standards. All *in vivo* study protocols were reviewed and approved by the Institutional Animal Care and Use Committee. Mice were either pretreated with vehicle, argatroban or *N*-acylpyrazole **73** prior to thrombin challenge. While survival in the group pre-treated with vehicle was 0%, the group pre-treated with argatroban survived in a dose dependent manner (see Table 12). All the mice pre-treated with *N*-acylpyrazole **73** at 9 mg/kg survived. While only a single dosage concentration of *N*-acylpyrazole **73** was tested in this assay, the 100% survival rate indicates that the *N*-acylpyrazole **73** did indeed act as an anticoagulant *in vivo*, approximately competitive with argatroban according to the limits of this assay.

Following these early *in vivo* results, and as the program progressed, we instituted more robust *in vivo* efficacy animal models. A fuller disclosure, and application of these more robust *in vivo* efficacy animal models, was recently disclosed by our laboratories for a compound with a much-improved PK profile versus those disclosed in this report [41]. Notwithstanding data from the undeniably more robust *in vivo* models, these results from the thrombin-induced thromboembolism model are the first to demonstrate a proof of concept for *N*-acylpyrazoles as a structural class.

2.4.2. *In vivo safety evaluation*

The tail bleeding time assay has been developed for the assessment of anticoagulants in rodents [55–57], and provides a useful surrogate for the clinical effect expected in man. Injury to the mouse is initiated by a tail transection, causing bleeding from the tail. In a 'normal', non-disease state, blood flow occurs for a period of time until clotting causes cessation of blood flow. In a 'non-normal', disease state (or a state that has been treated with an anticoagulant agent, a hemophilia agent, etc.), blood flow occurs for either a shorter or longer period of time, depending upon the effectiveness of all the factors in the coagulation cascade. The tail bleeding assay model has been refined to the point that bleeding time is well correlated with the animal species, the animal size, and the location of the cut, providing confidence in the robustness of this model in compound assessment. With regard to its relevance to the coagulation cascade, the model provides useful information on an anticoagulant's *in vivo* efficacy up to the point of a stable clot. It does not provide information on fibrinolysis i.e. clot dissolution.

Table 13
Bleeding study of argatroban, dabigatran and *N*-acylpyrazole **73** in CD-1 mice.

Study article	Tail bleeding time assay	
	Bleeding time (min)[f]	Blood cell count (x 10^6)/mL[f]
Vehicle[a]	9[d]	38[d]
Argatroban[b]	23 ± 1.5	380 ± 30
Dabigatran[c]	23 ± 1.5	250 ± 30
 **73**	25[e,g]	55 ± 20

[a] DMA:PEG300:phosphate buffer (5:1:4), plus 20 μM β-cyclodextrin.
[b] Dosed at 4 mg/kg in saline.
[c] Dosed at 0.2 mg/kg in saline.
[d] Standard deviation not determined.
[e] Dosed at 9 mg/kg in vehicle (DMA: PEG 300: Phosphate Buffer (5:1:4), plus 20 μM β-cyclodextrin).
[f] Standard deviation.
[g] Time limit of assay.

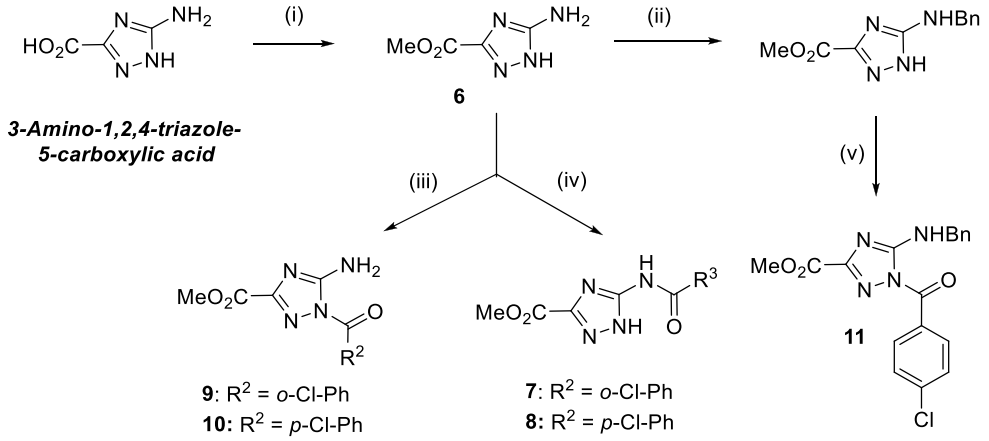
To that end, four study groups of CD-1 mice (male, 25–30 g) were acquired (Harlan Labs, Livermore, CA), prepared and handled according to AAALAC ethical standards. Mice were pretreated with either vehicle, argatroban, dabigatran or *N*-acylpyrazole **73**. Mice were subsequently anaesthetized and 2 mm of length of their tails transected and bleeding times and blood cell loss were measured. Bleeding time and total blood cell count from each study group are tabulated in Table 13.

Mice pre-treated with vehicle had an average bleeding time of only 9 min, with a blood cell count of 38 × 10^6 cells/mL. On the other hand, mice pre-treated with argatroban or dabigatran bled for an average of 24 and 25 min respectively, close to the end of the study time permitted. These bleeding times approached triple the bleeding time following the vehicle injection, and are in line with previous literature reports [58, 59]. Mice pre-treated with *N*-acylpyrazole **73** also bled for 25 min following transection; again, this provided a proof of concept that this class of compounds functioned *in vivo* as anticoagulants.

However, we were surprised by the stark difference between *N*-acylpyrazole **73** and the two positive controls, as it pertained to their blood cell counts following mouse tail transection. Pre-treatment with argatroban and dabigatran caused mice to lose 310 × 10^6 blood cells per mL and 250 × 10^6 blood cells per mL respectively, an increase of 6-8-fold vs. the vehicle. On the other hand, delivery of *N*-acylpyrazole **73** prior to tail transection led to an approximately 1.5-fold increase of liberated whole blood cells vs. that of the vehicle, a dramatic difference vs. blood cell count from treatment with the positive controls.

As an observation on these latter sections on *in vivo* studies, it would have been ideal to present a complete correlation between *in vivo* efficacy and safety results in mice, and *N*-acylpyrazole **73**'s *in vitro* inhibition data against both human and mouse thrombin isoforms. No murine thrombin inhibition assay data was, however, collected for *N*-acylpyrazole **73.** On the other hand, several measurements ***were*** collected on both argatroban and later compounds in this series (not reported in this manuscript); in every instance of our *N*-acylpyrazoles, inhibition of human thrombin differed (~2x) from that of murine thrombin by the same factor exhibited by that of argatroban (IC_{50} (human thrombin = 0.21 μM, vs. IC_{50} (mouse thrombin = 0.52 μM). The implication, although not rigorously proven, is that the human enzyme inhibition data and the human TGA data serve well as surrogates in correlating to mouse-based *in vivo* studies.

In closing, when translated from this *in vivo* test setting to a human setting, these *in vivo* results imply that desired anticoagulation using the known 'benchmark' DTIs will be accompanied by excessive bleeding. We speculate that this phenomenon plays a significant role in the undesired bleeding side-effect phenomena seen in all the reversible, active



Scheme 1. Synthesis of Triazoles **6**–**10** in Table 1 and Triazole **11** in Table 2.[a]
[a]Reagents and conditions: (i); $SOCl_2$, MeOH, 98%; (ii); PhCHO, EtOH, reflux, 6 h; cool to rt, $NaCNBH_3$, 16 h, rt, 15% over 2 steps; (iii); R^2COCl, py, rt, 21% (cmpd **9**), 37% (cmpd **10**); (iv); R^3COCl, Et_3N, 80 °C, 12% (cmpd **7**), 26% (cmpd **8**); (v); *p*-ClPhC(O)Cl, Et_3N, rt, 5 h, 47%.

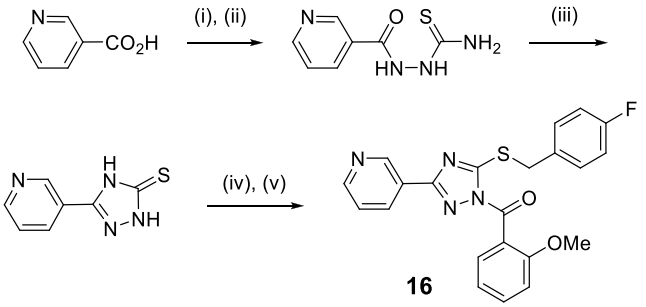
site thrombin inhibitors. This undesired phenomenon is effectively eliminated on treatment with the *N*-acylpyrazole **73**, and by extension all members of this motif as a structural class (data not shown). We speculate that because of the novelty of this structural class and the novelty of its mechanism of action vs. the mechanism of action of known, reversible, active site thrombin inhibitors, these *N*-acylpyrazoles offer a valid pathway to ideally safe and effective anticoagulants.

3. Chemistry

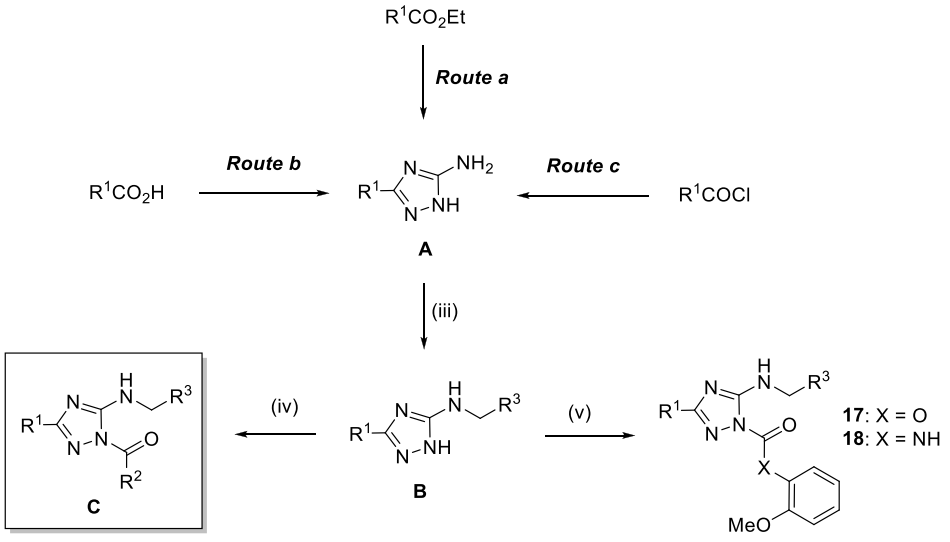
Synthesis of compounds in Table 1**.** Compounds **6**–**10** (Table 1) and **11** (Table 2) were prepared according to Scheme 1. Commercially available 3-amino-1,2,4-triazole-5-carboxylic acid was esterified using excess thionyl chloride to furnish methyl ester **6** almost quantitatively. Our amidation studies on methyl ester **6** paralleled observations from earlier investigators [60–64]; that is, dependent on the reaction conditions and reagent selection, we observed and quantified acylation at either N^{exo} or $N^{1\text{-}endo}$. Further explanation can be found in the Supporting Information.

Compound **11** was prepared from triazole **6** by reductive benzylation using sodium cyanoborohydride, followed by amidation of the $N^{1\text{-}endo}$ nitrogen in triethylamine at room temperature, as illustrated in Scheme 1 and exemplified in full in the Supporting Information.

General route for the synthesis of *N*-acyltriazoles. Compounds **12**, **15**–**22**, **28**–**47** were prepared according to the general route shown in Scheme 2. Starting from either carboxylic acids, esters, or acid chlorides, the core amino-1,2,4-triazole nucleus (**A**) was prepared *via* one of three general methods, each exemplified in the Supporting Information. Reductive amination conditions allowed selective extension from the exocyclic amine nitrogen [65,66], installing the R^3 side chain in the



Scheme 3. Synthesis of thioether **16**.[a]
[a]Reagents and conditions: (i); $(COCl)_2$, DCM, rt, then DMF, 2 h; (ii); thiosemicarbazide, py, 0 °C, then rt, 14 h; (iii); KOH, H_2O, reflux, 3 h (17% yield over 3 steps); (iv); *p*-F-PhCH$_2$Br, THF, H_2O, −10 °C, 8 h, 23%; (v); *o*-MeO-PhC(O)Cl, Et_3N, rt, 5 h, 30%.



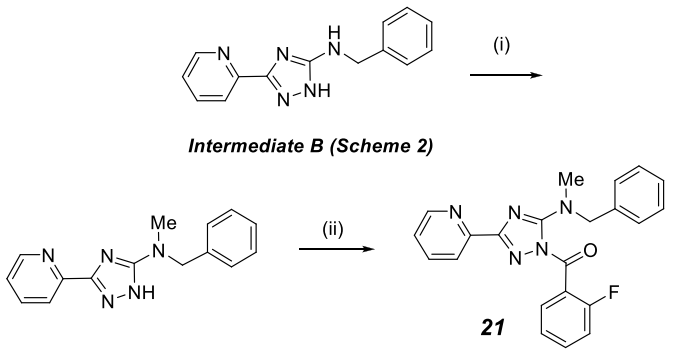
Scheme 2. General synthetic route towards *N*-acyltriazoles.[a]
[a]Reagents and conditions: ***Route a***; aminoguanidine sulfate, NaOMe, MeOH, 0 °C, then 65 °C, 20 h; ***Route b***; aminoguanidine sulfate, H_2SO_4, 140 °C, 72 h; ***Route c***; (i); aminoguanidine sulfate, py, 0 °C, then rt, 14 h; (ii); H_2O, reflux, 6 h; (iii); R^3CHO, EtOH, reflux, 6 h; cool to rt, $NaCNBH_3$, AcOH, 16 h, rt; (iv); $R^2C(O)Cl$, Et_3N, rt, 5 h; (v); (X = O): *o*-MeO-Ph-OC(O)Cl, Et_3N, rt, 5 h (**17**, 20% yield for step (v)); (X = NH): *o*-MeO-Ph-NCO, DMF, 0 °C, then rt, 6 h (**18**, 28% yield for step (v)).

general structure (**B**). Regioselective installation of the $N^{1\text{-}endo}$-acyl group then formed the *N*-acyltriazole (**C**) target compounds.

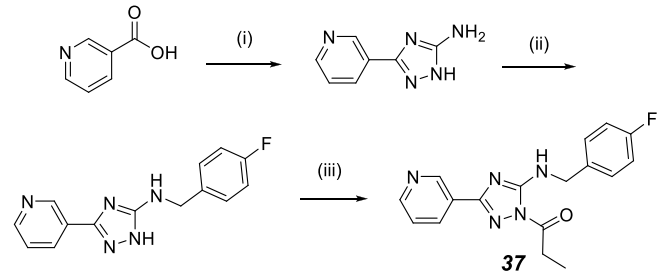
Synthesis of thioether 16 (Scheme 3). Thioether **16** was prepared in a similar fashion to ***route c*** in Scheme 2 [67]. As shown in Scheme 3, nicotinic acid was converted to the acid chloride hydrochloride salt on stirring with oxalyl chloride in the presence of DMF, and then converted to the intermediate hydrazide by amidation with thiosemicarbazide. This was cyclized under strongly basic conditions to the key 1,2,4-triazole-3-thione intermediate, which was then *S*-alkylated and finally *N*-acylated to furnish thioether **16**. Full experimental procedures for the synthesis of **16** are described in the Supporting Information.

Syntheses of carbamate 17 and urea 18 (Scheme 2). *N*-acylation with either the appropriate chloroformate or isocyanate, using the conditions outlined in Scheme 2 (step (v), both reagents prepared by standard means and used without further purification, furnished the carbamate **17** and urea **18** [61,68].

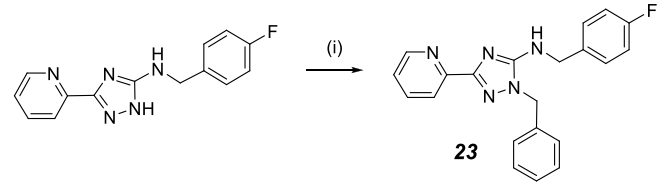
Synthesis of *N*-acyltriazole 21 (Scheme 4). The route employed for the synthesis of all *N*-acyltriazoles (*vide supra,* see Scheme 2) was intercepted by an additional exhaustive reductive amination step. Reductive amination of intermediate **B** with aqueous formaldehyde was followed by acylation of the $N^{1\text{-}endo}$-pyrazole nitrogen, as usual, to furnish the final product *N*-acyltriazole **21**.



Scheme 4. Synthesis of *N*-acyltriazole **21**.[a]
[a]Reagents and conditions: (i); HCHO (40% in H_2O), MeOH, AcOH, $NaCNBH_3$, 0 °C to rt, 4 h, 57%; (ii); *o*-F-PhC(O)Cl, Et_3N, CH_2Cl_2, rt, 4 h, 20%.



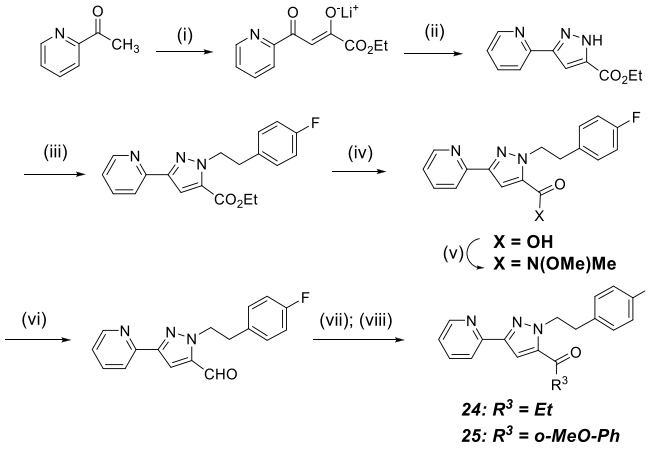
Scheme 5. Representative synthesis of *N*-acyltriazole **37**.[a]
[a]Reagents and conditions: (i); aminoguanidine sulfate, H_2SO_4, 140 °C, 72 h, 39%; (ii); *p*-F-Ph-CHO, EtOH, 4 Å MS, reflux, 8 h; cool to rt, $NaCNBH_3$, AcOH, 15 h, rt, 51%; (iii); EtC(O)Cl, Et_3N, rt, 5 h, 33%.



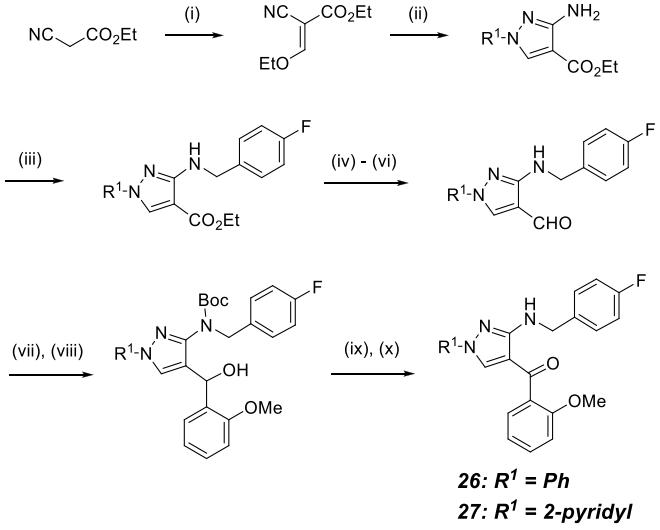
Scheme 6. Synthesis of N^{endo}-benzyltriazole **23**.[a]
[a]Reagents and conditions: (i); $PhCH_2Br$, K_2CO_3, DMF, rt, 5 h, 17%.

Representative synthesis of *N*-acyltriazole 37 (Scheme 5). A representative illustration of the general synthesis of *N*-acyltriazoles (Scheme 2) is shown in Scheme 5 for *N*-acyltriazole **37**, and the experimental procedure is outlined in full in the Supporting Information.

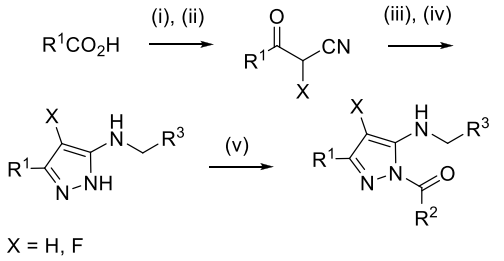
Synthesis of $N^{triazole}$-benzyl compound 23 (Scheme 6).



Scheme 7. Synthesis of pyrazole regioisomers **24** and **25**.[a]
[a]Reagents and conditions: (i); (a); LHMDS, THF, Et_2O, −78 °C, 30 min; (b); $(CO_2Et)_2$, Et_2O; (c); warm to rt, 20 h, 74%; (ii); $NH_2NH_2.H_2O$, AcOH, 100 °C, 12 h, 66%; (iii); *p*-F-Ph$(CH_2)_2$Br, K_2CO_3, DMF, rt, 8 h, 58%; (iv); LiOH.H_2O, THF, H_2O, MeOH, 50 °C, 3 h, 83%; (v); HN(OMe)Me.HCl, EDCI, HOBt, DIEA, DMF, rt, 16 h, 55%; (vi); $LiAlH_4$, THF, −40 °C, then 0 °C, 2 h, 79%; (vii); R^3MgCl, THF, −78 °C, then 0 °C, 4 h 84% (cmpd 24); (viii); MnO_2, THF, rt, 14 h (21% (cmpd 24); 60% over 2 steps (cmpd 25)).



Scheme 8. Synthesis of pyrazole regioisomers **26** and **27**.[a]
[a]Reagents and conditions: (i); $(EtO)_3CH$, Ac_2O, 140 °C, 5 h, 76%; (ii); R^1NHNH_2, NaOAc, AcOH, H_2O, 110 °C, 16 h (39% (R^1 = Ph); 38% (R^1 = 2-pyridyl)); (iii); *p*-F-PhCH_2Br, NaH, DMF, rt, 5 h (65% (R^1 = Ph); 55% (R^1 = 2-pyridyl)); (iv); KOH, MeOH, THF, 70 °C, 16 h (88% (R^1 = Ph); 84% (R^1 = 2-pyridyl)); (v); HN(OMe)Me.HCl, EDCI, HOBt, DIEA, DMF, rt, 16 h (61% (R^1 = Ph); 65% (R^1 = 2-pyridyl)); (vi); $LiAlH_4$, THF, −40 °C, then 0 °C, 5 h (~75% (R^1 = Ph); 80% (R^1 = 2-pyridyl)); (vii); Boc_2O, DMAP, Et_3N, THF, rt, 16 h (53% (R^1 = Ph); 76% (R^1 = 2-pyridyl)); (viii); *o*-MeO-PhMgBr, THF, −78 °C, then 0 °C, 4 h (70% (R^1 = Ph); 65% (R^1 = 2-pyridyl)); (ix); MnO_2, THF, rt, 48 h, (78% (R^1 = Ph); 75% (R^1 = 2-pyridyl)); (x); TFA, CH_2Cl_2, 0 °C, then rt, 2 h, (66% (R^1 = Ph); 73% (R^1 = 2-pyridyl)).



Scheme 9. General scheme for the synthesis of *N*-acylpyrazoles **48**–**91**.[a]
[a]Reagents and conditions: (i); $SOCl_2$, EtOH, 78 °C; (ii); X–CH_2-CN, base; (iii); $H_2NNH_2.H_2O$, EtOH, 78 °C (iv); R^3CHO, EtOH, 78 °C, then $NaCNBH_3$, AcOH, rt; (v); R^2COCl, Et_3N.

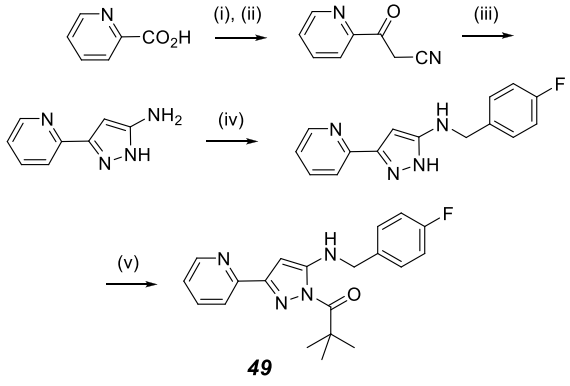
Interception of the N–*H* triazole intermediate with a benzyl group followed the chemistry illustrated in Scheme 6. The synthetic procedure is outlined in the Supporting Information.

Synthesis of pyrazole regioisomers 24 and 25 (Scheme 7). Pyrazole regioisomers **24** and **25** were synthesized in 8 steps through the sequence shown in Scheme 7. Installation of the appropriate R^3 group was achieved in the penultimate step of the sequence. The specific synthetic procedures are outlined in full in the Supporting Information.

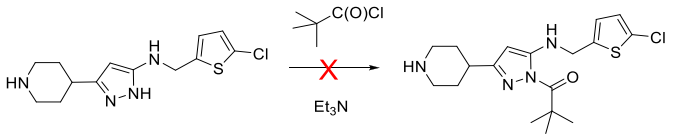
Synthesis of pyrazole regioisomers 26 and 27 (Scheme 8). Pyrazole regioisomers **26** and **27** were synthesized in 10 steps through the sequence shown in Scheme 8. Installation of the appropriate R^1 group was achieved in the 2nd step of the sequence. The specific synthetic procedures are outlined in the Supporting Information.

General route for the synthesis of *N*-acylpyrazoles (Scheme 9). Compounds **48**–**91** were all prepared according to the general route shown in Scheme 9. Carboxylic acids were converted to the ethyl esters, and thence to the α-(cyanomethyl)ketones by displacement with acetonitrile anion. These were difficult to purify and characterization revealed them to be a mixture of keto/enol isomers. Thus, the crude isolates were used 'as is' in the next reaction. Reaction with hydrazine led to the desired aminopyrazole scaffold through cyclocondensation, which could be easily purified. Reductive amination, as similarly conducted in the aminotriazole series, led to aminopyrazoles with the desired R^3 side chain. Finally, regioselective installation of the $N^{1\text{-}endo}$-acyl group formed *N*-acylpyrazole target compounds.

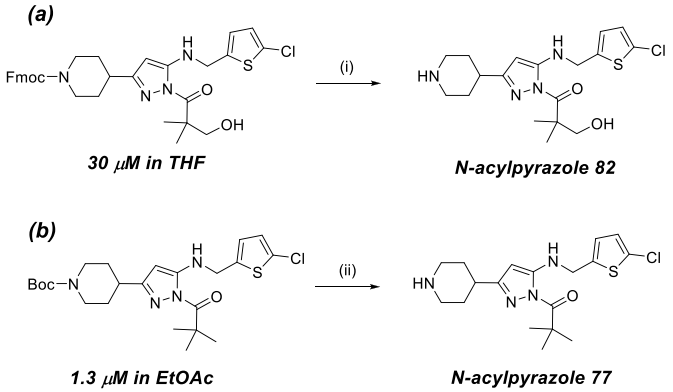
Representative synthesis of *N*-acylpyrazole 49 (Scheme 10). A representative illustration of the general synthesis (Scheme 9) of *N*-acylpyrazoles is shown in Scheme 10 for *N*-acylpyrazole **49**, and is outlined in full in the Supporting Information.



Scheme 10. Synthesis route to *N*-acylpyrazole **49**.[a]
[a]Reagents and conditions: (i); $SOCl_2$, EtOH, 78 °C, 81%; (ii); NaH, MeCN, PhMe, 65 °C, 16 h, 53%; (iii); $H_2NNH_2.H_2O$, EtOH, 78 °C, 20 h, 64% (iv); (a); *p*-F-PhCHO, EtOH, 4 Å MS, 78 °C, 8h; (b); AcOH, $NaCNBH_3$, 0 °C; (c) rt, 15 h, 67% (v); ^{t}BuCOCl, Et_3N, 3 h, 30%.



Scheme 11. Non-productive synthesis strategy for the preferred amidation of pyrazole N^{endo}-H vs. distal nucleophilic amines.

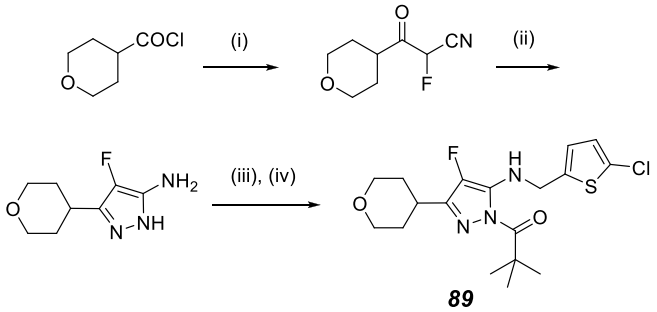


Scheme 12. Example carbamate deprotection reactions in the presence of *N*-acylpyrazole motifs.[a]
[a]Reagents and conditions: (i); piperidine, THF, rt, 30 min, 48%; (ii); Me_3SiI, EtOAc, 0 °C to rt, 15 min, 10%.

Synthetic strategies for the preparation of amine-substituted *N*-acylpyrazoles. The preparation of cyclic amines **76**–**79**, **82**, **83** faced synthetic challenges alluded to by the subject matter of this entire publication: since the pharmacological utility of these *N*-acylpyrazoles depended upon the electrophilic lability of the $N^{1\text{-}endo}$-acyl group, it became challenging to develop practical routes to unprotected amines in the presence of the sensitive $N^{1\text{-}endo}$-acyl group. On the one hand, synthesis strategies dictated that installation of the $N^{1\text{-}endo}$-acyl group be conducted as late as possible in any route. On the other hand, all attempts to acylate the pyrazole at the $N^{1\text{-}endo}$ nitrogen in preference to a distal amine failed (Scheme 11) due to the greater nucleophilicity of an aliphatic amine vs. a pyrazole amine [69].

With that in mind, the desired cyclic amines **76**–**79**, **82**, **83** were obtained by extensive experimentation and careful reaction monitoring, albeit in low yields. Procedures in Scheme 12 to prepare *N*-acylpyrazoles **77** and **82** are illustrative, the key being that deprotection of both the acid-sensitive Boc and the base-sensitive Fmoc groups were achieved under very dilute conditions. Experimental procedures for the preparation of **77** and **82** are described in full in the Supporting Information.

General route for the synthesis of *N*-acyl(4-fluoropyrazoles)



Scheme 13. Synthesis route to *N*-acylpyrazole **89**.[a]
[a]Reagents and conditions: (i); FCH_2CN, LHMDS, THF, −78 °C; (ii); $H_2NNH_2.H_2O$, EtOH, reflux; (iii); (a); 5-chloro-2-thiophenecarboxaldehyde, EtOH, 4 Å MS, reflux, then (b); $NaCNBH_3$, AcOH; (iv); ^{t}BuCOCl, Et_3N, rt, 5 h, 46%.

(Scheme 13**).** Compounds **84**–**86**, **88**, **89**, **91** were all prepared according to the general route shown in Scheme 9. However, taking into account peculiar reactivity considerations for fluoroacetonitrile, *N*-acyl (4-fluoropyrazoles) **84**–**86**, **88**, **89**, **91** were prepared according to specific synthesis adaptations, illustrated in Scheme 13 for compound **89** and exemplified in full in the Supporting Information.

4. Conclusions

We describe the discovery and profiling of a novel series of *N*-acylpyrazoles, which act as selective, covalent, reversible, non-competitive inhibitors of thrombin [40,41]. Chemical- and biological stabilities were optimized to provide compounds with *in vivo* efficacy albeit modest PK exposure. We report that, unlike current standards-of-care active site thrombin- and fXa inhibitor anticoagulants, these compounds do not inhibit platelet activation. Taken together, we suggest that covalent *N*-acylpyrazole inhibitors offer a unique and useful addition to safe and efficacious future anticoagulants. Subsequent work on optimization of *in vivo* exposure, measuring against both argatroban and standards-of care such as dabigatran and apixaban, while maintaining efficacy and stability, has been conducted and an advanced candidate in this series was recently reported by our laboratories [41]. Further work towards the selection of a clinical candidate will be disclosed in future communications.

5. Experimental section

Synthetic chemistry deployed in the preparation of all the final compounds in this article follows literature precedent. We describe salient details as follows for the sake of completeness, and to illustrate challenges that were evident in this project. ^{1}H NMR spectra and other appropriate data for all final compounds are available in the Supporting Information.

5.1. General chemistry information

FT-IR spectra were measured with a PerkinElmer Spectrum 100 FT-IR spectrometer using a KBr disc. Proton nuclear magnetic resonance (^{1}H NMR) spectra were measured with Varian MERCURYplus (400 MHz), Varian MERCURYplus (300 MHz), Bruker AvanceCore (400 MHz), Bruker Avance Neo (400 MHz) spectrometers. Chemical shifts are given in parts per million (ppm) with tetramethylsilane as an internal standard, and coupling constants (J values) are given in hertz (Hz). Splitting patterns and apparent multiplicities are designated as singlet (s), doublet (d), double doublet (dd), triplet (t), multiplet (m), and broad singlet (br s). Analytical thin-layer chromatography (TLC) was performed on silica gel 60 F_{254} plates (Merck). Column chromatography was carried out using silica gel (70–230 mesh, Merck). Liquid chromatography mass spectrometry (LCMS) analysis was performed on a Waters Acquity UPLC with SQD mass detector, operating in electrospray ionization (ESI) (+ or −) or atmospheric pressure chemical ionization (APCI) (+ or −) ionization mode. Analytes were eluted using a linear gradient of 0.025% TFA containing water/acetonitrile, or 0.05% formic acid containing water/acetonitrile, or 10 mM ammonium formate containing water/acetonitrile mobile phase and detected at 220 nm. Analytical high-performance liquid chromatography (HPLC) was performed on a Waters Alliance 2695 HPLC System with a Waters 2998 Photodiode Array (PDA) Detector. Waters Sunfire C-18 (150 mm × 4.6 mm I.D., 5 μm), or Waters Acquity BEH C-18 (100 mm × 2.1 mm. I.D., 1.7 μm), or YMC-Triart C18 (150 mm × 4.6 mm, I.D. 5 μm, VWR America), or Waters XBridge BEH C18 (150 mm × 4.6 mm, I.D. 5 μm) were used with a temperature of 30 °C and a flow rate of 0.3 mL/min. Mobile phases A and B under neutral condition were a mixture of 0.01% ammonium acetate in water, and acetonitrile. Mobile phases A and B under acidic conditions were either a mixture of 0.025% TFA in water, and 0.025% TFA in acetonitrile, respectively, or a mixture of 0.1% formic acid in water, and 0.1% formic acid in acetonitrile. All compounds are >95% pure by HPLC analysis. All commercially available solvents and reagents were used without further purification, unless otherwise stated. Yields were not optimized.

5.2. Spectral characterization of novel compounds

Compound **6** (Scheme 1)**:** ^{1}H NMR (DMSO-d_6, 400 MHz): δ 12.62 (s, 1H), 6.23 (s, 2H), 3.76 (s, 3H); IR: 3426 cm^{-1} (N–H); 3066 cm^{-1} (C–H); 1721 cm^{-1} (C═O); 1655 cm^{-1} (C═N); 1234 cm^{-1} (C–O); 1066 cm^{-1} (C–N) cm^{-1}; MP: 240–241 °C; LC-MS *m/z* 142.0 (RT = 3.44 min, 99.6%); HPLC purity: 99.4%, RT 3.45 min.

Compound **7** (Scheme 1)**:** ^{1}H NMR (DMSO-d_6, 400 MHz): δ 14.36 (s, 1H, D_2O exchangeable), 12.37 (s, 1H, D_2O exchangeable), 7.46–7.67 (m, 4H), 3.84 (s, 3H); MP: 100–101 °C; LC-MS *m/z* 280.96 (M + H$^+$) (RT = 0.99 min, 97.2%); HPLC purity: 97.5%, RT 1.55 min.

Compound **8** (Scheme 1)**:** ^{1}H NMR (DMSO-d_6, 400 MHz): δ 14.29 (br s, 1H, D_2O exchangeable), 12.37 (br s, 1H, D_2O exchangeable); 8.09 (d, *J* = 8.3 Hz, 2H), 7.64 (d, *J* = 8.3 Hz, 2H), 3.85 (s, 3H); IR: 3325 cm^{-1} (N–H); 3092 cm^{-1} (C–H); 1745 cm^{-1} (C═O); 1584 cm^{-1} (C═N); 1486 cm^{-1} (C═C); 1225 cm^{-1} (C–O); 1119 cm^{-1} (C–N) cm^{-1}; MP: 253–257 °C; LC-MS *m/z* 280.96 (M + H$^+$) (RT = 1.48 min, 99.0%); HPLC purity: 99.8%, RT 1.13 min.

Compound **9** (Scheme 1)**:** ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.00 (br s, 2H, D_2O exchangeable), 7.49–7.74 (m, 4H), 3.77 (s, 3H); MP: 238–241 °C; LC-MS *m/z* 281.02 (M + H$^+$) (RT = 1.69 min, 98.1%); HPLC purity: 98.6%, RT 1.25 min.

Compound **10** (Scheme 1)**:** ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.04 (d, *J* = 8.6 Hz, 2H), 7.89 (br s, 2H, D_2O exchangeable), 7.66 (d, *J* = 8.6 Hz, 2H), 3.82 (s, 3H); MP: 208–211 °C; LC-MS *m/z* 280.96 (M + H$^+$) (RT = 1.43 min, 98.2%); HPLC purity: 98.3%, RT 1.50 min.

Compound **11** (Scheme 1)**:** ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.56 (t, *J* = 6.2 Hz, 1H), 8.06 (d, *J* = 8.4 Hz, 2H), 7.67 (d, *J* = 8.4 Hz, 2H), 7.24–7.41 (m, 5H), 4.67 (d, *J* = 6.2 Hz, 2H), 3.81 (s, 3H); TLC: 20% EtOAc in hexane: R_f: 0.50. LC-MS *m/z* 371.06 (M + H$^+$) (RT = 2.25 min, 97.4%); HPLC purity: 97.6%, RT 3.15 min.

Compound **12** (Scheme 2)**:** ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.50 (t, *J* = 6.5 Hz, 1H), 7.81–7.84 (m, 2H), 7.08–7.56 (m, 11H), 4.67 (d, *J* = 6.5 Hz, 2H), 3.77 (s, 3H); MP: 130–133 °C; LC-MS *m/z* 403.09 (M + H$^+$) (RT = 1.59 min, 99.6%); HPLC purity: 99.8%, RT 2.52 min.

Compound **15** (Scheme 2)**:** ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.99 (s, 1H), 8.56–8.64 (m, 2H), 8.13 (d, *J* = 7.9 Hz, 1H), 7.07–7.59 (m, 9H), 4.68 (d, *J* = 6.1 Hz, 2H), 3.77 (s, 3H); MP: 136–139 °C; LC-MS *m/z* 404.11 (M + H$^+$) (RT = 3.26 min, 96.7%); HPLC purity: 97.1%, RT 3.10 min.

Compound **16** (Scheme 3)**:** ^{1}H NMR (DMSO-d_6, 400 MHz): δ 9.13 (s, 1H), 8.71 (d, *J* = 3.9 Hz, 1H), 8.26 (d, *J* = 7.9 Hz, 1H), 7.53–7.67 (m, 5H), 7.09–7.25 (m, 4H), 4.64 (s, 2H), 3.75 (s, 3H); MP: 108–112 °C; LC-MS *m/z* 421.08 (M + H$^+$) (RT = 1.39 min, 99.0%); HPLC purity: 99.0%, RT 2.65 min.

Compound **17** (Scheme 2): ^{1}H NMR (CD_3OD, 400 MHz): δ 9.19 (s, 1H), 8.58 (d, *J* = 3.9 Hz, 1H), 8.45 (d, *J* = 7.5 Hz, 1H), 7.52–7.59 (m, 3H), 6.95–7.33 (m, 7H), 5.42 (br s, 2H), 3.79 (s, 3H); LC-MS *m/z* 420.10 (M + H$^+$) (RT = 1.66 min, 94.0%); HPLC purity: 94.1%, RT 2.93 min.

Compound **18** (Scheme 2)**:** ^{1}H NMR (DMSO-d_6, 400 MHz): δ 9.40 (s, 1H), 9.20 (s, 1H), 8.69 (d, *J* = 4.0 Hz, 1H), 8.34 (d, *J* = 7.9 Hz, 1H), 8.19 (s, 1H), 7.96 (d, *J* = 7.9 Hz, 1H), 7.49–7.56 (m, 3H), 7.00–7.19 (m, 5H), 4.64 (d, *J* = 5.7 Hz, 2H), 3.93 (s, 3H); MP: 136–139 °C; LC-MS *m/z* 419.10 (M + H$^+$) (RT = 2.64 min, 98.4%); HPLC purity: 98.4%, RT 2.45 min.

Compound **19** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.98 (s, 1H), 8.62 (d, J = 4.4 Hz, 1H), 8.51 (br s, 1H), 8.12 (d, J = 7.9 Hz, 1H), 7.07–7.59 (m, 10H), 4.72 (d, J = 6.1 Hz, 2H), 3.78 (s, 3H); MP: 145–147 °C; LC-MS m/z 386.11 (M + H$^+$) (RT = 2.02 min, 98.9%); HPLC purity: 99.3%, RT 1.76 min.

Compound **20** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.55–8.61 (m, 2H), 7.80–7.97 (m, 3H), 7.69–7.71 (m, 1H), 7.35–7.49 (m, 7H), 7.28 (t, J = 7.3 Hz, 1H), 4.73 (d, J = 6.3 Hz, 2H); MP: 155–159 °C; LC-MS m/z 374.21 (M + H$^+$) (RT = 3.26 min, 97.8%); HPLC purity: 98.2%, RT 2.22 min.

Compound **21** (Scheme 4): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.61 (d, J = 3.8 Hz, 1H), 7.71–7.96 (m, 4H), 7.30–7.47 (m, 8H), 4.83 (s, 2H); 3.05 (s, 3H); MP: 106–108 °C; LC-MS m/z 388.18 (M + H$^+$) (RT = 2.06 min, 98.4%); HPLC purity: 99.3%, RT 2.53 min.

Compound **22** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.66 (t, J = 4.4 Hz, 1H), 8.55 (t, J = 6.0 Hz, 1H), 7.90–8.16 (m, 4H), 7.46–7.72 (m, 6H), 7.17 (t, J = 8.8 Hz, 2H), 4.72 (d, J = 6.0 Hz, 2H); MP: 130–132 °C; LC-MS m/z 374.08 (M + H$^+$) (RT = 1.98 min, 96.1%); HPLC purity: 96.7%, RT 2.32 min.

Compound **22a** (Fig. 5): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 13.1 (br s, D_2O exchangeable, 1H), 7.85–7.93 (m, 2H), 7.38–7.42 (m, 3H), 7.12–7.17 (m, 2H), 6.8 (br s, D_2O exchangeable, 1H), 4.37 (d, J = 6.2 Hz, 2H); MP: 219–220 °C; LC-MS m/z 270.08 (M + H$^+$) (RT = 1.14 min, 97.9%); HPLC purity: 98.2%, RT 1.38 min.

Compound **23** (Scheme 6): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.55 (m, 1H), 7.91 (d, J – 7.9 Hz, 1H), 7.81 (t, J = 7.5 Hz, 1H), 5.26 (s, 2H), 4.54 (d, J = 5.6 Hz, 2H); LC-MS m/z 360.11 (M + H$^+$) (RT = 1.66 min, 98.9%); HPLC purity: 99.1%, RT 1.04 min.

Compound **24** (Scheme 7): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.61 (d, J = 5.1 Hz, 1H), 7.85–7.95 (m, 2H), 7.60 (s, 1H), 7.35–7.38 (m, 1H), 7.08–7.21 (m, 4H), 4.74 (t, J = 7.6 Hz, 2H), 3.06 (t, J = 7.3 Hz, 2H), 2.94 (q, J = 7.2 Hz, 2H), 1.03 (t, J = 7.3 Hz, 3H); MS: 324 [M + H]$^+$; MP: 129–130 °C; TLC: 50% EtOAc in hexane: Rf: 0.50. LC-MS m/z 324.18 (M + H$^+$) (RT = 1.72 min, 98.4%); HPLC purity: 99.2%, RT 3.41 min.

Compound **25** (Scheme 7): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.53 (d, J = 4.7 Hz, 1H), 7.83–7.93 (m, 2H), 7.57 (dt, J = 8.0, 1.7 Hz, 1H), 7.06–7.35 (m, 8H), 6.84 (s, 1H); 4.84 (t, J = 7.3 Hz, 2H); 3.74 (s, 2H), 3.16 (t, J = 7.3 Hz, 2H); MS: 324 [M + H]+; MP: 129–133 °C; TLC: 50% EtOAc in hexane: Rf: 0.50. LC-MS m/z 402.21 (M + H$^+$) (RT = 3.19 min, 99.3%); HPLC purity: 99.4%, RT 3.52 min.

Compound **26** (Scheme 8): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 7.94 (t, J = 6.7 Hz, 1H), 6.96-7.51 (m, 14H), 4.11 (d, J = 6.7 Hz, 2H), 3.78 (s, 3H); MP: 140-141°C; LC-MS m/z 402.22 (M + H$^+$) (RT = 2.36 min, 98.3%); HPLC purity: 98.8%, RT 2.59 min.

Compound **27** (Scheme 8): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 9.72-9.74 (m, 1H), 8.49 (br d, J = 3.8 Hz, 1H), 8.01 (t, J = 6.9 Hz, 1H), 7.82 (d, J = 8.2 Hz, 1H), 7.00-7.48 (m, 10H), 4.88 (d, J = 6.9 Hz, 2H), 3.71 (s, 3H); MP: 82-84°C; LC-MS m/z 403.16 (M + H$^+$) (RT = 2.41 min, 99.7%); HPLC purity: 99.2%, RT 3.49 min.

Compound **28** (Scheme 2): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 8.58 (d, J = 4.4 Hz, 1H), 8.47 (t, J = 6.1 Hz, 1H), 7.85-7.94 (m, 2H), 7.21-7.59 (m, 9H), 7.09 (t, J = 7.0 Hz, 1H), 4.72 (d, J = 6.1 Hz, 2H), 3.77 (s, 3H); MP: 185-188°C; LC-MS m/z 386.11 (M + H$^+$) (RT = 3.56 min, 98.4%); HPLC purity: 98.7%, RT 2.06 min.

Compound **28a** (Fig. 5): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 13.5 (br s, D_2O exchangeable, 1H), 12.4 (br s, D_2O exchangeable, 1H), 8.60 (dd, J = 27.2, 4.4 Hz, 2H), 7.8-8.0 (m, 2H), 7.2-7.45 (m, 6H), 6.8 (br s, D_2O exchangeable, 1H), 4.40 (dd, J = 29.0, 6.1 Hz, 2H); IR: 3305 cm^{-1}; 1591 cm^{-1}; 1550 cm^{-1}; 746 cm^{-1}; MP: 226-228°C; LC-MS m/z 252.09 (M + H$^+$) (RT = 1.12 min, 99.2%); HPLC purity: 99.1%, RT 2.35 min.

Compound **29** (Scheme 2): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 8.58-8.62 (m, 2H), 7.78-7.95 (m, 3H), 7.26-7.66 (m, 9H), 4.74 (d, J = 6.1 Hz, 2H); MP: 182-186°C; LC-MS m/z 390.07 (M + H$^+$) (RT = 2.78 min, 99.1%); HPLC purity: 99.7%, RT 2.51 min.

Compound **30** (Scheme 2): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 8.65 (d, J = 5.5 Hz, 1H), 8.60 (t, J = 6.1 Hz, 1H), 7.89-8.09 (m, 4H), 7.46-7.49 (m, 3H), 7.25-7.38 (m, 3H), 4.74 (d, J = 6.1 Hz, 2H); MP: 153-155°C; LC-MS m/z 381.20 (M + H$^+$) (RT = 2.20 min, 97.9%); HPLC purity: 98.2%, RT 1.49 min.

Compound **31** (Scheme 2): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 8.58 (d, J = 4.4 Hz, 1H), 8.50 (t, J = 6.1 Hz, 1H), 7.85-7.94 (m, 2H), 7.42-7.58 (m, 5H), 7.07-7.23 (m, 4H), 4.68 (d, J = 6.1 Hz, 2H), 3.77 (s, 3H); MP: 158-161°C; LC-MS m/z 404.11 (M + H$^+$) (RT = 3.62 min, 98.4%); HPLC purity: 98.9%, RT 2.12 min.

Compound **32** (Scheme 2): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 8.59 (t, J = 6.0 Hz, 1H), 7.99-8.02 (m, 2H), 7.71-7.73 (m, 2H), 7.41-7.53 (m, 5H), 7.26-7.29 (m, 1H), 7.18 (d, J = 2.7 Hz, 1H), 6.99 (dd, J = 4.8, 3.7 Hz, 1H), 4.87 (d, J = 6.0 Hz, 2H), 3.84 (s, 3H); MP: 149-151°C; LC-MS m/z 391.20 (M + H$^+$) (RT = 1.95 min, 99.9%); HPLC purity: 99.8%, RT 2.56 min.

Compound **33** (Scheme 2): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 8.59 (t, J = 6.0 Hz, 1H), 7.99-8.02 (m, 2H), 7.71-7.73 (m, 2H), 7.41-7.53 (m, 5H), 7.26-7.29 (m, 1H), 7.18 (d, J = 2.7 Hz, 1H), 6.99 (dd, J = 4.8, 3.7 Hz, 1H), 4.75 (d, J = 6.1 Hz, 2H), 3.77 (s, 3H); MP: 144-145°C; LC-MS m/z 425.07 (M + H$^+$) (RT = 4.08 min, 98.7%); HPLC purity: 99.1%, RT 3.13 min.

Compound **34** (Scheme 2): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 8.60 (d, J = 4.7 Hz, 1H), 8.51 (t, J = 6.0 Hz, 1H), 7.99 (d, J = 7.6 Hz, 1H), 7.90 (t, J = 7.6 Hz, 1H), 7.42-7.59 (m, 4H), 7.17-7.23 (m, 2H), 7.09 (t, J = 7.6 Hz, 1H), 7.00 (t, J = 4.3 Hz, 1H), 4.86 (d, J = 6.0 Hz, 2H), 3.76 (s, 3H); MP: 206-207°C; LC-MS m/z 392.17 (M + H$^+$) (RT = 2.04 min, 99.5%); HPLC purity: 99.7%, RT 1.93 min.

Compound **35** (Scheme 2): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 8.61 (d, J = 4.4 Hz, 1H), 8.55 (t, J = 6.1 Hz, 1H), 8.00 (d, J = 7.9 Hz, 1H), 7.91 (dt, J = 7.6, 1.5 Hz, 1H), 7.44-7.59 (m, 3H), 7.22 (d, J = 8.4 Hz, 1H), 7.09 (t, J = 7.5 Hz, 1H), 7.03 (d, J = 3.5 Hz, 1H), 7.00 (d, J = 3.5 Hz, 1H), 4.77 (d, J = 6.1 Hz, 2H), 3.77 (s, 3H); MP: 145-148°C; LC-MS m/z 426.22 (M + H$^+$) (RT = 2.10 min, 97.6%); HPLC purity: 98.8%, RT 1.35 min.

Compound **36** (Scheme 2): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 8.66 (d, J = 4.4 Hz, 1H), 8.26 (t, J = 6.1 Hz, 1H), 8.03 (d, J = 7.9 Hz, 1H), 7.89-7.93 (m, 1H), 7.23-7.48 (m, 6H), 4.67 (d, J = 6.1 Hz, 2H), 3.06 (q, J = 7.5 Hz, 2H); 1.16 (t, J = 7.5 Hz, 3H); IR: 3393 cm^{-1} (N–H); 2931 cm^{-1} (C–H); 1717 cm^{-1} (C═O); 1605 cm^{-1} (C═N); 1529 cm^{-1} (C═C); 1151 cm^{-1} (C–N) cm^{-1}; MP: 116-118°C; LC-MS m/z 308.07 (M + H$^+$) (RT = 3.84 min, 98.1%); HPLC purity: 98.2%, RT 2.91 min.

Compound **37** (Scheme 5): ^{1}H NMR (DMSO-d$_6$, 400 MHz): δ 9.14 (d, J = 1.7 Hz, 1H), 8.66 (dd, J = 4.8, 1.3 Hz, 1H), 8.28–8.34 (m, 2H, 1H D_2O exchangeable), 7.47–7.53 (m, 3H), 7.15 (t, J = 8.8 Hz, 2H), 4.63 (d, J = 6.1 Hz, 2H), 3.05 (q, J = 7.4 Hz, 2H); 1.16 (t, J = 7.4 Hz, 3H); MP: 144–146 °C; LC-MS m/z 326.07 (M + H$^+$) (RT = 1.89 min, 98.5%); HPLC purity: 98.9%, RT 2.81 min.

Compound **38** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.65 (d, J = 4.1 Hz, 1H), 8.26 (br s, 1H), 8.02 (d, J = 7.7 Hz, 1H), 7.88–7.92 (m, 1H), 7.44–7.47 (m, 3H), 7.14 (t, J = 8.9 Hz, 2H), 4.63 (d, J = 6.1 Hz, 2H), 3.05 (q, J = 7.2 Hz, 2H); 1.16 (t, J = 7.2 Hz, 3H); MP: 98–99 °C; LC-MS m/z 326.01 (M + H$^+$) (RT = 1.92 min, 96.4%); HPLC purity: 97.3%, RT 1.28 min.

Compound **39** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.67 (d, J = 4.4 Hz, 1H), 8.29 (br s, 1H), 8.04 (d, J = 7.9 Hz, 1H), 7.92 (t, J = 7.4 Hz, 1H), 7.45–7.50 (m, 3H), 7.16 (t, J = 8.8 Hz, 2H), 4.64 (d, J = 6.1 Hz, 2H), 3.61–3.68 (m, 1H); 1.23 (d, J = 7.0 Hz, 6H); MP: 116–118 °C; LC-MS m/z 340.10 (M + H$^+$) (RT = 1.90 min, 98.0%); HPLC purity: 98.9%, RT 2.18 min.

Compound **40** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.68 (d, J = 4.1 Hz, 1H), 8.40 (br s, 1H), 8.04 (d, J = 7.8 Hz, 1H), 7.91 (t, J = 7.4 Hz, 1H), 7.44–7.48 (m, 3H), 7.15 (t, J = 8.7 Hz, 2H), 4.64 (d, J = 6.1 Hz, 2H), 1.47 (s, 9H); MP: 73–76 °C; LC-MS m/z 354.01 (M + H$^+$) (RT = 2.12 min, 99.1%); HPLC purity: 99.8%, RT 1.71 min.

Compound **41** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.69 (d, J = 4.4 Hz, 1H), 8.28 (t, J = 6.2 Hz, 1H), 8.06 (d, J = 7.9 Hz, 1H), 7.93 (t, J = 7.9 Hz, 1H), 7.47–7.50 (m, 1H), 7.36 (d, J = 8.3 Hz, 2H), 6.90 (d, J =

8.3 Hz, 2H), 4.59 (d, J = 6.2 Hz, 2H), 3.72 (s, 3H), 1.47 (s, 9H); MP: 119–121 °C; LC-MS *m/z* 366.16 (M + H^+) (RT = 3.85 min, 95.0%); HPLC purity: 95.7%, RT 2.41 min.

Compound **42** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.25 (t, J = 6.1 Hz, 1H), 7.26–7.52 (m, 7H), 7.15 (d, J = 8.8 Hz, 1H), 7.03 (t, J = 7.4 Hz, 1H), 4.60 (d, J = 6.1 Hz, 2H), 3.76 (s, 3H), 2.01 (s, 3H); MP: 136–137 °C; LC-MS *m/z* 323.12 (M + H^+) (RT = 3.04 min, 98.9%); HPLC purity: 99.6%, RT 3.30 min.

Compound **43** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.25 (t, J = 6.1 Hz, 1H), 7.27–7.54 (m, 7H), 7.16 (d, J = 8.4 Hz, 1H), 7.04 (t, J = 7.5 Hz, 1H), 4.60 (d, J = 6.1 Hz, 2H), 3.76 (s, 3H), 2.66–2.68 (m, 1H), 1.10 (d, J = 7.0 Hz, 6H); MP: 134–136 °C; LC-MS *m/z* 351.22 (M + H^+) (RT = 2.39 min, 98.8%); HPLC purity: 98.9%, RT 1.76 min.

Compound **44** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.17 (t, J = 6.1 Hz, 1H), 7.25–7.54 (m, 7H), 7.16 (d, J = 8.4 Hz, 1H), 7.04 (t, J = 7.5 Hz, 1H), 4.60 (d, J = 6.1 Hz, 2H), 3.75 (s, 3H), 2.36–2.42 (m, 1H); 1.58–1.78 (m, 5H); 1.14–1.36 (m, 5H); MP: 110–114 °C; LC-MS *m/z* 391.07 (M + H^+) (RT = 1.76 min, 99.6%); HPLC purity: 99.8%, RT 2.38 min.

Compound **45** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.28 (t, J = 6.2 Hz, 1H), 7.97 (ddd, J = 7.6, 7.6, 1.6 Hz, 1H), 7.23–7.53 (m, 8H), 4.64 (d, J = 6.2 Hz, 2H), 3.02 (q, J = 7.2 Hz, 2H); 1.14 (t, J = 7.2 Hz, 3H); MP: 113–114 °C; LC-MS *m/z* 325.2 (M + H^+) (RT = 2.64 min, 98.7%); HPLC purity: 98.3%, RT 2.74 min.

Compound **46** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.47 (t, J = 6.2 Hz, 1H), 7.93 (ddd, J = 7.6, 7.6, 1.4 Hz, 1H), 7.16–7.57 (m, 8H), 7.06 (t, J = 7.4 Hz, 1H), 7.00 (dd, J = 4.8, 3.4 Hz, 1H), 4.84 (d, J = 6.2 Hz, 2H), 3.77 (s, 3H); MP: 187–189 °C; LC-MS *m/z* 409.16 (M + H^+) (RT = 1.43 min, 99.6%); HPLC purity: 98.7%, RT 2.23 min.

Compound **47** (Scheme 2): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 9.10 (br s, 1H), 8.60 (m, 1H), 8.15 (t, J = 7.2 Hz, 1H), 7.91 (d, J = 4.8 Hz, 1H), 7.73–7.75 (m, 1H), 7.55–7.60 (m, 1H), 7.36–7.42 (m, 3H), 7.17 (br s, 1H), 6.99 (dd, J = 4.4, 4.0 Hz, 1H), 4.86 (d, J = 6.2 Hz, 2H); MP: 107–110 °C; LC-MS *m/z* 385.07 (M + H^+) (RT = 2.91 min, 99.8%); HPLC purity: 99.0%, RT 3.02 min.

Compound **48** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.58 (br s, 1H), 7.89 (br s, 1H), 7.77 (t, J = 7.3 Hz, 1H), 7.47–7.62 (m, 3H), 7.36 (t, J = 5.8 Hz, 1H), 7.19 (d, J = 8.1 Hz, 1H), 7.01–7.09 (m, 3H), 6.08 (s, 1H), 4.60 (d, J = 6.1 Hz, 2H), 3.75 (s, 3H); IR: 3394 cm^{-1} (N–H); 2959 cm^{-1} (C–H); 1685 cm^{-1} (C═O); 1594 cm^{-1} (C═N); 1493 cm^{-1} (C═C); 1249 cm^{-1} (C–O); 1160 cm^{-1} (C–N) cm^{-1}; 761 cm^{-1} (C–Cl); MP: 151–153 °C; LC-MS *m/z* 425.01 (M + H^+) (RT = 3.31 min, 99.4%); HPLC purity: 99.7%, RT 1.77 min.

Compound **49** (Scheme 10): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.58 (d, J = 4.4 Hz, 1H), 7.86-7.98 (m, 3H), 7.38-7.46 (m, 3H), 7.18 (t, J = 8.8 Hz, 2H), 5.84 (s, 1H), 4.40 (d, J = 6.2 Hz, 2H), 1.50 (s, 9H); MS: 353 [M + H]+; MP: 102-103°C; TLC: 20% EtOAc in hexane: R_f: 0.60 ^{1}H NMR: (DMSO-d_6) δ 8.58 (d, J = 4.4 Hz, 1H), 7.85-7.98 (m, 3H), 7.38-7.46 (m, 3H), 7.18 (t, J = 8.8 Hz, 2H), 5.84 (s, 1H), 4.40 (d, J = 6.1 Hz, 2H), 1.50 (s, 9H); MP: 102-103°C; LC-MS m/z 353.19 (M + H^+) (RT = 2.39 min, 99.7%); HPLC purity: 100.0%, RT 3.27 min.

Compound **50** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 7.79-7.84 (m, 3H), 7.37-7.49 (m, 5H), 7.17 (t, J = 8.8 Hz, 2H), 5.89 (s, 1H), 4.37 (d, J = 6.2 Hz, 2H), 1.49 (s, 9H); LC-MS m/z 352.15 (M + H^+) (RT = 3.20 min, 98.7%); HPLC purity: 99.2%, RT 3.89 min.

Compound **51** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.99 (d, J = 1.5 Hz, 1H), 8.58 (dd, J = 5.3, 1.2 Hz, 1H), 8.13-8.16 (m, 1H), 7.83 (t, J = 6.2 Hz, 1H), 7.45-7.48 (m, 3H), 7.16 (t, J = 8.7 Hz, 2H), 6.01 (s, 1H), 4.38 (d, J = 6.2 Hz, 2H), 1.49 (s, 9H); LC-MS m/z 353.17 (M + H^+) (RT = 1.78 min, 98.3%); HPLC purity: 98.4%, RT 1.53 min.

Compound **52** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 9.21 (s, 1H), 8.87 (d, J = 4.8 Hz, 1H), 7.93-7.97 (m, 2H), 7.43-7.47 (m, 2H), 7.17 (t, J = 8.8 Hz, 2H), 5.93 (s, 1H), 4.41 (d, J = 6.2 Hz, 2H), 1.50 (s, 9H); MP: 155-157°C; LC-MS m/z 354.23 (M + H^+) (RT = 2.60 min, 99.9%); HPLC purity: 99.8%, RT 2.18 min.

Compound **53** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.89 (d, J = 4.8 Hz, 2H), 7.85 (br s, 1H), 7.43-7.51 (m, 3H), 7.18 (t, J = 8.8 Hz, 2H), 5.90 (s, 1H), 4.40 (d, J = 6.1 Hz, 2H), 1.50 (s, 9H); MP: 154-156°C; LC-MS m/z 354.10 (M + H^+) (RT = 2.39 min, 99.8%); HPLC purity: 99.9%, RT 1.80 min.

Compound **54** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 9.46 (d, J = 4.8 Hz, 1H), 8.67 (br s, 1H), 8.34-8.37 (m, 1H), 7.99 (br s, 1H), 7.44 (br s, 2H), 7.17 (t, J = 8.8 Hz, 2H), 5.95 (s, 1H), 4.47 (br s, 2H), 1.51 (s, 9H); MP: 154-155°C; LC-MS m/z 354.3 (M + H^+) (RT = 3.99 min, 99.8%); HPLC purity: 98.7%, RT 3.71 min.

Compound **55** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.58 (br s, 1H), 7.86-7.98 (m, 3H), 7.25-7.41 (m, 6H), 5.82 (s, 1H), 4.41 (d, J = 6.0 Hz, 2H), 1.51 (s, 9H); MP: 79-82°C; LC-MS m/z 335.20 (M + H^+) (RT = 2.98 min, 97.8%); HPLC purity: 99.4%, RT 3.23 min.

Compound **56** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.63 (d, J = 4.7 Hz, 1H), 7.99 (d, J = 7.6 Hz, 1H), 7.89 (ddd, J = 7.6, 7.6, 1.5 Hz, 1H), 7.61-7.66 (m, 2H), 7.42 (dd, J = 7.3, 5.5 Hz, 1H), 6.41 (br s, 2H), 6.05 (s, 1H), 4.41 (d, J = 6.0 Hz, 2H), 1.50 (s, 9H); MP: 99-101°C; LC-MS m/z 325.12 (M + H^+) (RT = 2.10 min, 99.0%); HPLC purity: 99.9%, RT 3.36 min.

Compound **57** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.61 (d, J = 4.7 Hz, 1H), 7.81-7.99 (m, 3H), 7.39-7.42 (m, 2H), 7.12 (d, J = 2.8 Hz, 1H), 6.99 (dd, J = 4.7, 3.5 Hz, 1H), 6.00 (s, 1H), 4.60 (d, J = 6.0 Hz, 2H), 1.50 (s, 9H); MP: 88-90°C; LC-MS m/z 341.14 (M + H^+) (RT = 2.23 min, 99.5%); HPLC purity: 99.7%, RT 2.50 min.

Compound **58** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.63 (br s, 1H), 7.86-8.02 (m, 2H), 7.76 (br s, 1H), 7.40-7.54 (m, 3H), 7.17 (br s, 1H), 5.97 (s, 1H), 4.42 (br s, 2H), 1.53 (s, 9H); MP: 91-93°C; LC-MS m/z 341.20 (M + H^+) (RT = 2.26 min, 97.2%); HPLC purity: 98.4%, RT 2.49 min.

Compound **59** (Scheme 9): ^{1}H NMR ($CDCl_3$, 400 MHz): δ 8.63 (br s, 1H), 7.86-8.02 (m, 2H), 7.76 (br s, 1H), 7.40-7.54 (m, 3H), 7.17 (br s, 1H), 6.27 (s, 1H), 4.49 (d, J = 5.8 Hz, 2H), 1.57 (s, 3H), 1.54 (s, 6H); MP: 154-157°C; LC-MS m/z 376.19 (M + H^+) (RT = 3.43 min, 98.9%); HPLC purity: 99.0%, RT 2.66 min.

Compound **60** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.61 (d, J = 4.4 Hz, 1H), 7.86-7.99 (m, 3H), 7.41 (br s, 1H), 6.96-7.02 (m, 2H), 6.01 (s, 1H), 4.54 (d, J = 5.8 Hz, 2H), 1.50 (s, 9H); MP: 98-100°C; LC-MS m/z 375.06 (M + H^+) (RT = 2.13 min, 99.6%); HPLC purity: 99.6%, RT 3.97 min.

Compound **61** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.58 (d, J = 4.8 Hz, 1H), 7.85 (t, J = 6.1 Hz, 1H), 7.80 (ddd, J = 7.9, 7.9, 1.3 Hz, 1H), 7.69 (d, J = 7.9 Hz, 1H), 7.35-7.38 (m, 1H), 7.04-7.07 (m, 3H), 7.00 (d, J = 3.5 Hz, 1H), 6.94 (t, J = 7.7 Hz, 1H), 6.08 (s, 1H), 4.60 (d, J = 6.1 Hz, 2H), 4.27 (d, J = 2.6 Hz, 2H), 4.20 (d, J = 2.6 Hz, 2H); MP: 172-175°C; LC-MS m/z 452.95 (M + H^+) (RT = 3.39 min, 99.0%); HPLC purity: 98.9%, RT 2.79 min.

Compound **62** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 9.02 (s, 1 H), 8.63 (d, J = 4.8 Hz, 1H), 8.19 (d, J = 7.9 Hz, 1H), 7.97 (t, J = 6.1 Hz, 1H), 7.87-7.91 (m, 2H), 7.43 (dd, J = 6.6, 5.8 Hz, 1H), 7.11 (d, J = 1.8 Hz, 1H), 7.04 (d, J = 4.0 Hz, 1H), 6.98 (d, J = 4.0 Hz, 1H), 6.13 (s, 1H), 4.61 (d, J = 6.1 Hz, 2H); MP: 133-135°C; LC-MS m/z 384.99 (M + H^+) (RT = 3.54 min, 98.8%); HPLC purity: 98.7%, RT 3.14 min.

Compound 63 (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 9.10 (br s, 1 H), 8.63 (d, J = 4.8 Hz, 1H), 8.11 (d, J = 7.9 Hz, 1H), 7.99 (br t, J = 6.1 Hz, 1H), 7.86-7.92 (m, 2H), 7.70 (dd, J = 5.3, 3.1 Hz, 1H), 7.43 (dd, J = 6.8, 5.5 Hz, 1H), 7.05 (d, J = 3.6 Hz, 1H), 6.98 (d, J = 3.6 Hz, 1H), 6.14 (s, 1H), 4.61 (d, J = 5.7 Hz, 2H); MP: 140-142°C; LC-MS m/z 400.97 (M + H^+) (RT = 3.67 min, 96.8%); HPLC purity: 96.6%, RT 3.44 min.

Compound **64** (Scheme 9): ^{1}H NMR ($CDCl_3$, 400 MHz): δ 8.60 (d, J = 4.0 Hz, 1H), 7.88 (d, J = 7.9 Hz, 1H), 7.65 (ddd, J = 7.9, 7.9, 1.8 Hz, 1H), 7.59 (d, J = 8.3 Hz, 1H), 7.46 (br t, J = 5.3 Hz, 1H), 7.20-7.24 (m, 1H), 6.85 (d, J = 3.5 Hz, 1H), 6.77 (d, J = 3.5 Hz, 1H), 6.54-6.58 (m, 2H), 6.11 (s, 1H), 4.53 (d, J = 5.3 Hz, 2H), 3.89 (s, 3H), 3.81 (s, 3H); MP: 125-127°C; LC-MS m/z 455.06 (M + H^+) (RT = 1.48 min, 95.9%); HPLC purity: 96.1%, RT 2.84 min.

Compound **65** (Scheme 9): ^{1}H NMR ($CDCl_3$, 400 MHz): δ 8.60 (d, J = 4.4 Hz, 1H), 7.74-7.86 (m, 3H), 7.69 (t, J = 6.1 Hz, 1H), 7.36-7.40 (m, 1H), 7.27 (t, J = 8.3 Hz, 1H), 7.05 (d, J = 3.5 Hz, 1H), 6.99 (d, J = 3.5 Hz, 1H), 6.83 (d, J = 8.3 Hz, 1H), 6.59 (t, J = 7.4 Hz, 1H), 6.24 (br s, 1H), 6.12 (s, 1H), 4.59 (d, J = 6.1 Hz, 2H); MP: 137-139°C; LC-MS m/z 410.11 (M + H^+) (RT = 1.37 min, 99.1%); HPLC purity: 99.7%, RT 2.65 min.

Compound **66** (Scheme 9): ^{1}H NMR (DMSO-d_6, 300 MHz): δ 8.61 (d, J = 4.6 Hz, 1H), 8.07 (d, J = 7.3 Hz, 2H), 7.96 (t, J = 5.7 Hz, 1H), 7.84-7.85 (m, 2H), 7.55-7.70 (m, 3H), 7.40 (dd, J = 8.5, 4.4 Hz, 1H), 7.07 (d, J = 3.5 Hz, 1H), 6.99 (d, J = 3.4 Hz, 1H), 6.15 (s, 1H), 4.62 (d, J = 5.7 Hz, 2H); MP: 128-130°C; LC-MS m/z 395.12 (M + H^+) (RT = 1.74 min, 99.6%); HPLC purity: 99.6%, RT 1.69 min.

Compound **67** (Scheme 9): ^{1}H NMR ($CDCl_3$, 400 MHz): δ 8.58 (d, J = 4.1 Hz, 1H), 7.87 (d, J = 8.0 Hz, 1H), 7.61-7.68 (m, 2H), 7.45 (ddd, J = 7.1, 7.1, 1.6 Hz, 1H), 7.33 (t, J = 5.6 Hz, 1H), 7.17-7.26 (m, 4H), 6.94-7.05 (m, 4H), 6.82 (d, J = 3.6 Hz, 1H), 6.75 (d, J = 3.6 Hz, 1H), 6.07 (s, 1H), 4.49 (d, J = 6.0 Hz, 2H); MP: 138-140°C; LC-MS m/z 487.22 (M + H^+) (RT = 2.52 min, 99.0%); HPLC purity: 99.2%, RT 2.43 min.

Compound **68** (Scheme 9): ^{1}H NMR ($CDCl_3$, 400 MHz): δ 8.64 (d, J = 4.8 Hz, 1H), 7.97 (d, J = 8.0 Hz, 1H), 7.75 (ddd, J = 7.8, 7.8, 1.6 Hz, 1H), 7.49 (br s, 1H), 7.29-7.32 (m, 1H), 6.83 (d, J = 3.6 Hz, 1H), 6.77 (d, J = 3.7 Hz, 1H), 6.07 (s, 1H), 4.49 (d, J = 5.7 Hz, 2H), 4.20 (t, J = 6.6 Hz, 1H), 3.84 (d, J = 6.6 Hz, 2H), 1.57 (s, 3H), 1.53 (s, 3H); LC-MS m/z 391.22 (M + H^+) (RT = 1.87 min, 96.1%); HPLC purity: 95.8%, RT 3.31 min.

Compound **69** (Scheme 9): ^{1}H NMR ($CDCl_3$, 400 MHz): δ 8.63 (d, J = 3.9 Hz, 1H), 8.04 (d, J = 7.8 Hz, 1H), 7.74 (t, J = 7.5, 1H), 7.48 (t, J = 5.4 Hz, 1H), 7.28-7.29 (m, 1H), 6.81 (d, J = 3.9 Hz, 1H), 6.75 (d, J = 3.9 Hz, 1H), 6.03 (s, 1H), 4.46 (d, J = 5.9 Hz, 2H), 3.92 (s, 2H), 3.28 (s, 3H), 1.52 (s, 6H); LC-MS m/z 405.22 (M + H^+) (RT = 2.65 min, 95.5%); HPLC purity: 95.4%, RT 2.08 min.

Compound **70** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.61 (d, J = 4.3 Hz, 1H), 8.01 (d, J = 8.0 Hz, 1H), 7.88 (ddd, J = 7.6, 7.6, 1.9 Hz, 1H), 7.80 (t, J = 6.0 Hz, 1H), 7.41 (ddd, J = 7.4, 4.8, 1.0 Hz, 1H), 7.02 (d, J = 3.6 Hz, 1H), 6.97 (d, J = 3.7 Hz, 1H), 6.06 (s, 1H), 4.55 (d, J = 5.9 Hz, 2H), 3.80-3.96 (m, 3H), 3.48 (ddd, J = 11.6, 11.6, 2.0 Hz, 2H), 1.65-1.93 (m, 4H); MP: 120-122°C; LC-MS m/z 403.24 (M + H^+) (RT = 1.95 min, 99.1%); HPLC purity: 99.8%, RT 3.44 min.

Compound **71** (Scheme 9): ^{1}H NMR ($CDCl_3$, 400 MHz): δ 8.63 (d, J = 4.3 Hz, 1H), 8.02 (d, J = 7.9 Hz, 1H), 7.75 (ddd, J = 7.6, 7.6, 1.0 Hz, 1H), 7.47-7.52 (m, 1H), 7.27-7.29 (m, 1H), 6.83 (d, J = 3.1 Hz, 1H), 6.77 (d, J = 4.0 Hz, 1H), 6.05 (s, 1H), 4.49 (d, J = 7.8 Hz, 2H), 3.80-3.85 (m, 2H), 3.66 (d, J = 11.7, 9.2, 2.6 Hz, 2H), 2.62 (d, J = 14.1 Hz, 2H), 1.84 (ddd, J = 14.1, 9.2, 3.5 Hz, 2H), 1.67 (m, 3H); MP: 108-110°C; LC-MS m/z 417.24 (M + H^+) (RT = 1.56 min, 98.5%); HPLC purity: 98.2%, RT 1.83 min.

Compound **72** (Scheme 9): ^{1}H NMR ($CDCl_3$, 400 MHz): δ 8.62 (d, J = 4.0 Hz, 1H), 8.05 (d, J = 7.5 Hz, 1H), 7.74 (t, J = 7.7 Hz, 1H), 7.50 (br s, 1H), 7.27-7.28 (m, 1H), 6.82 (d, J = 3.5 Hz, 1H), 6.76 (d, J = 3.9 Hz, 1H), 6.03 (s, 1H), 4.47 (d, J = 5.7 Hz, 2H), 2.50-2.55 (m, 2H), 1.40-1.61 (m, 11H); LC-MS m/z 415.26 (M + H^+) (RT = 2.82 min, 99.5%); HPLC purity: 99.3%, RT 3.08 min.

Compound **73** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 11.86 (s, 1H), 8.09 (dd, J = 7.0, 2.0 Hz, 1H), 7.71 (t, J = 6.2 Hz, 1H), 7.47 (dd, J = 6.2, 2.0 Hz, 1H), 6.96 (d, J = 3.6 Hz, 1H), 6.93 (d, J = 3.6 Hz, 1H), 6.32 (t, J = 6.6 Hz, 1H), 6.22 (s, 1H), 4.47 (d, J = 6.2 Hz, 2H), 1.47 (s, 9H); MP: 205-208°C; LC-MS m/z 391.22 (M + H^+) (RT = 2.89 min, 99.0%); HPLC purity: 99.1%, RT 2.33 min.

Compound **74** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 11.38 (s, 1H), 7.93 (t, J = 6.0 Hz, 1H), 7.56 (t, J = 6.2 Hz, 1H), 7.02 (d, J = 3.6 Hz, 1H), 6.99 (d, J = 3.7 Hz, 1H), 6.46 (br s, 1H), 6.1 (br s, 1H), 4.47 (d, J = 6.0 Hz, 2H), 1.47 (s, 9H); LC-MS m/z 391.24 (M + H^+) (RT = 2.40 min, 97.9%); HPLC purity: 98.7%, RT 9.32 min.

Compound **75** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 11.7-12.0 (br s, 1H), 7.93 (t, J = 6.0 Hz, 1H), 7.84-7.90 (m, 3H), 7.04 (d, J = 3.7 Hz, 1H), 6.97 (d, J = 3.7 Hz, 1H), 6.45 (d, J = 10.7 Hz, 1H), 5.95 (s, 1H), 4.46 (d, J = 5.7 Hz, 2H), 1.45 (s, 9H); LC-MS m/z 391.28 (M + H^+) (RT = 2.36 min, 100.0%); HPLC purity: 95.5%, RT 8.95 min.

Compound **76** (trifluoroacetic acid salt) (Scheme 12b): ^{1}H NMR ($CDCl_3$, 400 MHz): δ 7.55 (t, J = 5.7 Hz, 1H), 6.80 (d, J = 3.9 Hz, 1H), 6.77 (d, J = 3.5 Hz, 1H), 5.40 (s, 1H), 4.79 (dd, J = 7.1 4.4 Hz, 1H), 4.38 (d, J = 5.7 Hz, 2H), 3.41-3.53 (m, 2H), 2.02-2.41 (m, 5H), 1.45 (s, 9H); LC-MS m/z 367.22 (M + H^+) (RT = 2.75 min, 96.3%); HPLC purity: 97.1%, RT 3.20 min.

Compound **77** (trifluoroacetic acid salt) (Scheme 12b): ^{1}H NMR ($CDCl_3$, 400 MHz): δ 9.3–9.7 (br s, 2H), 7.49 (t, J = 5.7 Hz, 1H), 6.76 (s, 2H), 5.15 (s, 1H), 4.35 (d, J = 5.7 Hz, 2H), 3.45–3.49 (m, 2H), 3.06–3.11 (m, 2H), 2.85–2.87 (m, 1H), 2.12–2.26 (m), 4H), 1.45 (s, 9H); LC-MS *m/z* 381.2 (M + H^+) (RT = 3.24 min, 99.8%); HPLC purity: 98.7%, RT 2.07 min.

Compound **78** (formic acid salt) (Scheme 12b): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.36 (br s, 1H), 7.68 (t, J = 6.4 Hz, D_2O exchangeable, 1H), 6.99 (d, J = 3.6 Hz, 1H), 6.97 (d, J = 3.6 Hz, 1H), 5.53 (s, 1H), 4.43 (d, J = 6.2 Hz, 2H), 3.08–3.12 (m, 2H), 2.80–2.86 (m, 2H), 2.08–2.12 (m, 2H), 1.63–1.68 (m, 2H), 1.41 (s, 9H); 1.19 (s, 3H); LC-MS *m/z* 395.4 (M + H^+) (RT = 2.18 min, 100.0%); HPLC purity: 98.4%, RT 6.63 min.

Compound **79** (formic acid salt) (Scheme 12b): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.26 (br s, 0.5H), 7.70 (m, 1H), 6.97 (s, 2H), 5.41 (s, 1H), 4.41 (d, J = 6.0 Hz, 2H), 2.0–3.3 (m, 9H), 1.41 (s, 9H); LC-MS *m/z* 381.5 (M + H^+) (RT = 2.17 min, 86.1%); HPLC purity: 95.2%, RT 6.50 min.

Compound **80** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 7.71 (m, 1H), 6.96 (s, 2H), 5.41 (s, 1H), 4.65–4.67 (m, 1H), 4.43 (d, J = 6.1 Hz, 2H), 3.71–3.83 (m, 2H), 1.89–2.12 (m, 4H), 1.40 (s, 9H); LC-MS *m/z* 368.20 (M + H^+) (RT = 3.04 min, 98.9%); HPLC purity: 99.2%, RT 1.40 min.

Compound **81** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 7.68 (t, J = 6.1 Hz, 1H), 6.98 (d. J = 3.5 Hz, 1H), 6.96 (d, J = 3.5 Hz, 1H), 5.40 (s, 1H), 4.41 (d, J = 6.1 Hz, 2H), 3.87 (dd, J = 7.9, 2.6 Hz, 2H), 3.41 (ddd, J = 11.4, 11.4, 1.7 Hz, 2H), 2.67–2.75 (m, 1H), 1.78 (dd, J = 13.0, 2.0 Hz, 2H), 1.54–1.64 (m, 2H), 1.40 (s, 9H); LC-MS *m/z* 382.26 (M + H^+) (RT = 3.08 min, 96.3%); HPLC purity: 97.3%, RT 3.59 min.

Compound **82** (trifluoroacetic acid salt) (Scheme 12a). TLC System: 10% Methanol in dichloromethane. R_f-0.15; ^{1}H NMR ($CDCl_3$, 400 MHz): δ 9.4–9.8 (br s, 2H), 7.49 (t, J = 5.7 Hz, 1H), 6.78 (s, 1H), 6.77 (s, 1H), 5.21 (s, 1H), 4.37 (d, J = 5.7 Hz, 2H), 4.01 (br s, 1H), 3.75 (d, J = 3.0 Hz, 2H), 3.48–3.52 (m, 2H), 3.02–3.08 (m, 2H), 2.82–2.88 (m, 1H), 2.12–2.23 (m, 4H), 1.43 (s, 6H); MP: 156–159 °C. LC-MS *m/z* 397.31 (M + H^+) (RT = 1.49 min, 98.1%); HPLC purity: 98.2%, RT 2.37 min.

Compound **83** (HCl salt) (Scheme 12b): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.2–8.8 (br m, 2H), 7.70 (t, J = 6.0 Hz, 1H), 7.47 (t, J = 7.9 Hz, 1H), 7.35 (d, J = 7.3 Hz, 1H), 7.12 (d, J = 8.4 Hz, 1H), 6.97–7.03 (m, 3H), 5.43 (s, 1H), 4.49 (d, J = 6.2 Hz, 2H), 3.73 (s, 3H), 3.20 (d, J = 12.4 Hz, 2H), 2.80–2.92 (m, 2H), 2.6–2.8 (m, 1H), 1.80–1.88 (m, 2H), 1.55–1.68 (m, 2H); MP: 234–236 °C; LC-MS *m/z* 431.25 (M + H^+) (RT = 1.64 min, 97.8%); HPLC purity: 98.3%, RT 2.70 min.

Compound **84** (trifluoroacetic acid salt) (Schemes 12(b), 13): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.60 (br m, 1H), 7.50 (t, J = 7.4 Hz, 2H), 7.38 (d, J = 7.4 Hz, 1H), 7.14 (d, J = 8.3 Hz, 1H), 6.99–7.04 (m, 2H), 6.90 (d, J = 3.5 Hz, 1H), 4.58 (d, J = 6.5 Hz, 2H), 3.73 (s, 3H), 3.19–3.22 (m, 2H), 2.88–2.94 (m, 3H), 1.68–1.88 (m, 4H); MP: 186–188 °C; LC-MS *m/z* 449.18 (M + H^+) (RT = 1.71 min, 99.0%); HPLC purity: 98.6%, RT 2.98 min.

Compound **85** (trifluoroacetic acid salt) (Schemes 12(b), 13): ^{1}H NMR ($CDCl_3$, 400 MHz): δ 9.7 (br s, 1H), 9.5 (br s, 1H), 7.10 (br s, 1H), 6.75 (br s, 2H), 4.55 (br s, 2H), 3.4–3.6 (m, 2H), 3.0–3.2 (m, 3H), 2.2–2.3 (m, 4H), 1.43 (s, 9H); MP: 61–65 °C; LC-MS *m/z* 399.07 (M + H^+) (RT = 2.05 min, 95.7%); HPLC purity: 97.5%, RT 2.16 min.

Compound **86** (Schemes 12(b), 13): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 8.8 (br s, 1H), 7.49 (t, J = 6.2 Hz, 1H), 6.96 (d, J = 3.1 Hz, 1H), 6.85 (br s, 1H), 4.88 (br s, 1H), 4.5–4.6 (br s, 2H), 3.8–3.85 (br s, 2H), 3.2–3.4 (m, 2H), 2.8–3.1 (m, 3H), 1.80–2.05 (m, 4H), 1.30 (s, 6H); LC-MS *m/z*

415.25 (M + H^+) (RT = 1.55 min, 92.2%); HPLC purity: 92.8%, RT 5.22 min.

Compound **87** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 7.66 (t, J = 6.2 Hz, D_2O exchangeable, 1H), 7.47 (dt, J = 7.9, 7.9, 1.7 Hz, 1H), 7.35 (dd, J = 7.4, 1.7 Hz, 1H), 7.12 (d, J = 8.3 Hz, 1H), 6.98–7.04 (m, 3H), 5.50 (s, 1H), 4.48 (d, J = 6.1 Hz, 2H), 3.79–3.82 (m, 2H), 3.73 (s, 3H), 3.28–3.34 (m, 2H), 2.54–2.62 (m, 1H), 1.44–1.61 (m, 4H); LC-MS *m/z* 432.27 (M + H^+) (RT = 2.04 min, 97.9%); HPLC purity: 98.5%, RT 2.36 min.

Compound **88** (Scheme 13): ^{19}F NMR (DMSO-d_6, 377 MHz): δ −185.00; ^{1}H NMR (DMSO-d_6, 400 MHz): δ 7.02 (ddd, J = 7.6, 7.6, 1.6 Hz, 1H), 7.42 (t, J = 7.2 Hz, D_2O exchangeable, 1H), 7.36 (dd, J = 7.6, 1.6 Hz, 1H), 7.13 (d, J = 8.4 Hz, 1H), 7.02 (ddd, J = 7.6, 7.6, 0.8 Hz, 1H), 6.99 (d, J = 3.6 Hz, 1H), 6.90 (d, J = 3.6 Hz, 1H), 4.57 (d, J = 7.2 Hz, 2H), 3.79–3.82 (m, 2H), 3.72 (s, 3H), 3.31–3.37 (m, 2H), 2.76–2.82 (m, 1H), 1.56–1.62 (m, 4H); LC-MS *m/z* 450.01 (M + H^+) (RT = 2.39 min, 98.0%); HPLC purity: 99.5%, RT 4.18 min.

Compound **89** (Scheme 13): ^{19}F NMR (DMSO-d_6, 377 MHz): δ −188.24; ^{1}H NMR (DMSO-d_6, 400 MHz): δ 7.43 (t, J = 6.8 Hz, D_2O exchangeable, 1H), 6.96 (d, J = 3.6 Hz, 1H), 6.85 (d, J = 3.6 Hz, 1H), 4.51 (d, J = 6.0 Hz, 2H), 3.86–3.91 (m, 2H), 3.44 (ddd, J = 11.6, 11.6, 2.4 Hz, 2H), 2.88–2.94 (m, 1H), 1.78–1.81 (m, 2H), 1.62–1.71 (m, 2H), 1.40 (s, 9H); LC-MS *m/z* 400.06 (M + H^+) (RT = 2.70 min, 96.0%); HPLC purity: 96.8%, RT 4.84 min.

Compound **90** (Scheme 9): ^{1}H NMR (DMSO-d_6, 400 MHz): δ 7.66 (t, J = 6.4 Hz, D_2O exchangeable, 1H), 6.95–6.97 (m, 2H), 5.39 (s, 1H), 4.80 (t, J = 5.6 Hz, 1H), 4.41 (d, J = 6.4 Hz, 2H), 3.85–3.88 (m, 4H), 3.41 (ddd, J = 11.6, 11.6, 2.4 Hz, 2H), 2.67–2.72 (m, 1H), 1.78 (dd, J = 13.2, 2.0 Hz, 2H), 1.57–1.64 (m, 2H), 1.31 (s, 6H); LC-MS *m/z* 398.56 (M + H^+) (RT = 2.21 min, 95.6%); HPLC purity: 96.4%, RT 3.76 min.

Compound **91** (Scheme 13): ^{19}F NMR (DMSO-d_6, 377 MHz): δ −188.62; ^{1}H NMR (DMSO-d_6, 400 MHz): δ 7.44 (t, J = 6.8 Hz, D_2O exchangeable, 1H), 6.96 (d, J = 3.6 Hz, 1H), 6.85 (d, J = 3.6 Hz, 1H), 4.84 (t, J = 5.6 Hz, 1H), 4.51 (d, J = 6.8 Hz, 2H), 3.83–3.90 (m, 4H), 3.44 (ddd, J = 11.6, 11.6, 2.0 Hz, 2H), 2.89–2.91 (m, 1H), 1.80 (br d, J = 12.0 Hz, 2H), 1.64–1.71 (m, 2H), 1.30 (s, 6H); LC-MS *m/z* 416.04 (M + H^+) (RT = 2.31 min, 99.8%); HPLC purity: 98.3%, RT 3.99 min.

5.3. *In vitro assays*

Enzyme inhibition. Studies assessing the inhibition of various serine proteases were performed as described in detail earlier [40,41]. Briefly, IC_{50} values were determined by first pre-incubating the enzyme (thrombin, fXa or trypsin) with the compound in assay buffer for 10 min. The substrate was added, and the enzyme activity measured based on an increase in absorbance at 405 nm after 10 min using either a Cytation 3 or Synergy H1 plate reader (Biotek). Final reaction volumes were 100 μL.

Prothrombin time (PT) and activated partial thromboplastin time (aPTT) clotting assays. The aPTT and PT assays were performed as previously described [40], and as recommended by the vendor (Diagnostica Stago). Briefly, citrated normal pooled human platelet-poor plasma was mixed with test compound (2% DMSO) at 37 °C. Subsequently, either the aPTT assay reagent or PT assay reagent was added and mixed thoroughly. $CaCl_2$ was then added to trigger the clotting process. The time for clot formation was measured in seconds and the test compound concentration to double the clotting time (CT_{2x}) was determined.

Thrombin generation assay (TGA). The TGA was performed as previously described [40,42]. Briefly, the assay was performed with human platelet-poor plasma (final concentration 70%), containing phospholipids and Dade Innovin. Thrombin generation was initiated by the addition of a solution containing substrate and $CaCl_2$. The reaction was monitored kinetically by fluorescence at 37 °C for 90 min. The data were analyzed by plotting fluorescence intensity versus time and converting to a thrombogram showing the amount of free thrombin produced (in nM) versus time.

Thrombin time assay (TT). The TT was performed as previously described [26], and as recommended by the vendor (Diagnostica Stago). Briefly, a serial dilution of test compounds in DMSO was added to citrated normal pooled human platelet-poor plasma and mixed thoroughly (2% final DMSO conc.). Following a 10-min incubation, STA-Thrombin reagent (Diagnostica Stago) was added to the reaction mixture to trigger the clotting process. The time for clot formation was measured in seconds and the test compound concentration to double the clotting time (CT_{2x}) was determined.

Platelet activation assay in platelet-rich plasma. This assay was performed as previously described [41]. Platelet-rich plasma (PRP) was prepared from fresh CD-1 mouse blood collected in citrate vials, then centrifuged at 100x relative centrifugal force (RCF) for 10 min at room temperature. The top layer was collected. The assay was performed by mixing the resultant PRP with the following: assay buffer (10 mM HEPES, 150 mM NaCl, 1 mM $MgCl_2$, 3 mM KCl, pH 7.4), Gly-Pro-Arg-Pro (GPRP) (to prevent clotting), 2% DMSO in assay buffer with test compound, and finally mouse thrombin (10 nM). The reaction was incubated for 2 min, then 10 μL of the reagent ChronoLume (Chrono-log Corp, Havertown, PA) was added and the luminescence read after incubation for another 5 min.

Platelet activation assay in whole human blood. This assay was performed as previously described [41]. Human whole blood were collected in citrate vials, and the platelet activation assay was performed with 76% whole blood in assay buffer with GPRP and DMSO (2% in buffer). DMSO was diluted before addition to the blood to prevent hemolysis. 76% Whole blood mimics *in vivo* conditions as closely as possible. Test compounds diluted in assay buffer, were added to the blood samples just prior to platelet activation. Thrombin (final concentration 10 nM) was added for a total assay volume of 100 μL. For activation with tissue factor, Dade Innovin (Siemens, Tarrytown, NY) at a final dilution of 1/5000 and calcium chloride at 10 mM (final) were used. Immediately, 2 μL anti-CD61 (APC) and 2 μL anti-CD62P (PE) were added, mixed thoroughly, and incubated for 15 min. The samples were then diluted 100 × in PBS and then immediately read in a flow cytometer (CytoFlex, Beckman Coulter, Brea, CA). Total platelet population was identified using the CD61 antibody and the percentage of activated platelets was determined using the CD62P antibody.

Kinetics assay. A solution of compound **19** (500 μL, 0–2.88 nM) in 1,3-*bis*(*tris*(hydroxymethyl)methylamino)propane (BTP; 10 mM pH 7.4 in 145 mM NaCl, 0.1% PEG 8000) was incubated with a solution of human α-thrombin (200 μl, 0.4 nM) for 30 min at 37 °C. The reaction was started by adding a solution of the thrombin-specific chromogenic substrate (D)-FPR-*p*NA (100 μl) at a concentration of 8, 16, 48, 96 and 160 μM. Release of *p*NA resulting from thrombin-mediated (D)-FPR-*p*NA cleavage was monitored at 405 nm in a Cary-50 spectrophotometer (Varian) equipped with a temperature control system. Data analysis and visualization was carried out using Origin software.

5.4. *In vivo assays*

PK studies. Pharmacokinetics (PK) studies in mice were performed as described previously [40]. Test compound was dissolved in 90% Solutol HS15 and 10% ethanol. Mice were dosed orally with a flexible oral gavage needle at time t = 0. Plasma samples were collected at 0.25, 0.5, 1, 2, 4, 8, and 24 h post dose. The compound was extracted using acetonitrile, and analyzed by LC-MS to determine concentration. PK parameters were calculated using Phoenix WinNonlin software (Certara).

***In vivo* efficacy studies.** All *in vivo* study protocols were reviewed and approved by the Institutional Animal Care and Use Committee.

Thrombin-induced thromboembolism. Thrombin-induced thromboembolism studies in mice have been previously described [52]. For this study, mice were pre-treated with either vehicle alone, positive control (argatroban, 2, 4 or 8 mg/kg) or test article

N-acylpyrazole **73** (9 mg/kg in DMA: PEG 300: Phosphate Buffer (5:1:4), plus 20 mM β-cyclodextrin). All pre-treatment was done intravenously through the tail vein. One minute following pretreatment, human thrombin (1.5 U) was injected slowly into the tail vein. Mice in every group were monitored individually, and a record made of their mortality (i.e. death/survival) over 60 min.

Tail bleeding time test. The tail bleeding time assay has been developed for the assessment of anticoagulants in rodents [55–57]. For this study, tail bleeding time and blood cells loss were measured. Mice were divided into 4 study groups and each study group was pretreated with either vehicle (DMA: PEG 300: Phosphate Buffer (5:1:4), plus 20 μM β-cyclodextrin), argatroban (8 mg/kg (isotonic saline), dabigatran (0.2 mg/kg in saline) or *N*-acylpyrazole **73** in vehicle (DMA: PEG 300: Phosphate Buffer (5:1:4), plus 20 μM β-cyclodextrin). The pre-treatments were delivered intravenously through the tail vein. All mice were anaesthetized 1 min later with an appropriate dose of 2,2,2-tribromoethanol (300 mg/kg, diluted into *tert*-amyl alcohol and water (1:20 v/v). Each animal was examined for non-responsiveness (paw pinch), then a 2 mm length of the tail was transected. The animal was placed on a warming pad, then the transected tail was lowered into warm (37 °C) isotonic saline. A timer was started when the bleeding began, and stopped when the bleeding stopped. An important quality criterion was that cessation of bleeding had to be for >20 s; an important ethical criterion was that the overall study ceased at 25 min.

In addition to the bleeding time observations, the *total blood cells* from the exudate of each animal were counted. This was done by centrifugation of the collected blood/saline mixture from each animal, resuspension and dilution into isotonic saline, and inspection/counting of the number of cells using a hemocytometer.

Author contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Enrico DiCera reports financial support was provided by National Institutes of Health. Kevin M. Short has patent #US9,533,967 issued to Verseon Corporation. Kevin M. Short has patent #US9,963,440 issued to Verseon Corporation. Kevin M. Short has patent #US10,653,674 issued to Verseon Corporation.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ejmech.2022.114855.

Abbreviations

AAALAC	American Association for Accreditation of Laboratory Animal Care
aPC	activated protein C
aPTT	activated partial thromboplastin time
AUC	area under the curve
Boc	*tert*-butoxycarbonyl
Cl_{int}	intrinsic clearance
C_{max}	maximum concentration of drug
cmpd	compound
CT	clotting time
DCM	dichloromethane
(D)-FPR-*p*NA	D-Pro-Phe-Arg *p*-nitroanilide
DIEA	*N*,*N*-Diisopropylethylamine
DMA	*N*,*N*-dimethylacetamide
DMAP	4-Dimethylaminopyridine
DMF	dimethylformamide
DMSO	dimethyl sulfoxide
DOAC	Direct oral anticoagulant
DTI	direct thrombin inhibitor
EC_{50}	half maximal effective concentration
ECT	ecarin clotting time
EDCI	1-ethyl-3-(3-dimethylaminopropyl)carbodiimide
ETP	endogenous thrombin potential
FDA	United States Food and Drug Administration
fIXa	factor IXa
fXa	factor Xa
fXIa	factor XIa
Fmoc	fluorenylmethoxycarbonyl
FT-IR	Fourier-transform infrared spectroscopy
^1H	proton
GPRP	Gly-Pro-Arg-Pro
HEPES	(4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid)
HOBt	hydroxybenzotriazole
HPLC	High-performance liquid chromatography
IC_{50}	half maximal inhibitory concentration
INR	international normalized ratio
LHMDS	lithium hexamethyldisilazide
M	molar concentration
MP	Melting point
MS	molecular sieves
NMR	nuclear magnetic resonance
NS5A	Nonstructural protein 5A
PBS	Phosphate-buffered saline
PEG 300	Poly(ethylene glycol), mol wt 300
PEG 8000	Poly(ethylene glycol) average mol wt 8000
PK	pharmacokinetics
pNA	*p*-nitroaniline
ppm	parts per million
PRP	platelet-rich plasma
PT	prothrombin time
R_f	retention factor
rt	room temperature
SAR	structure-activity relationship
SD rat	Sprague Dawley rat
Ser	serine
$t_{½}$	half-life
TF	tissue factor
TFA	trifluoroacetic acid
TGA	thrombin generation assay
THF	tetrahydrofuran
TLC	thin layer chromatography
T_{max}	time to peak drug concentration
TT	thrombin time.

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